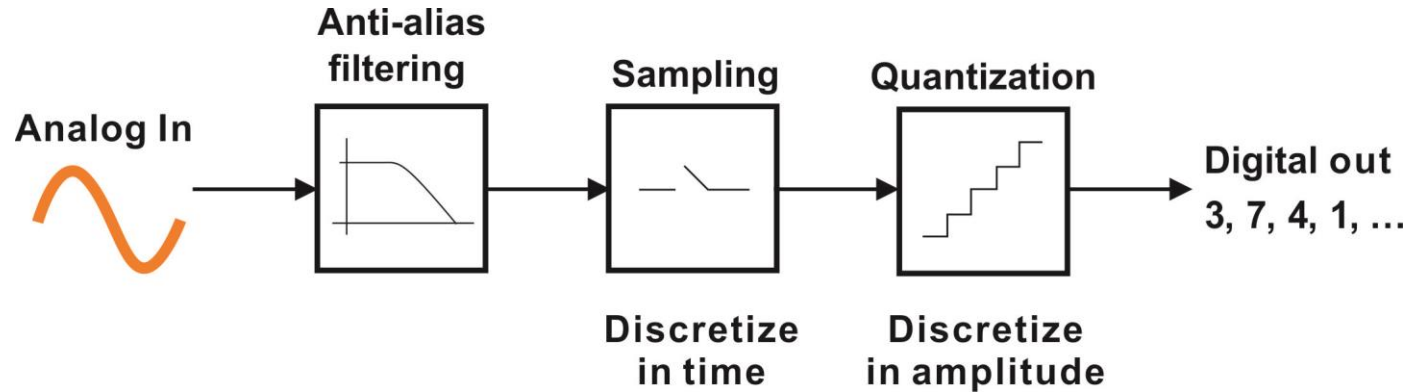


# Sampling Circuits

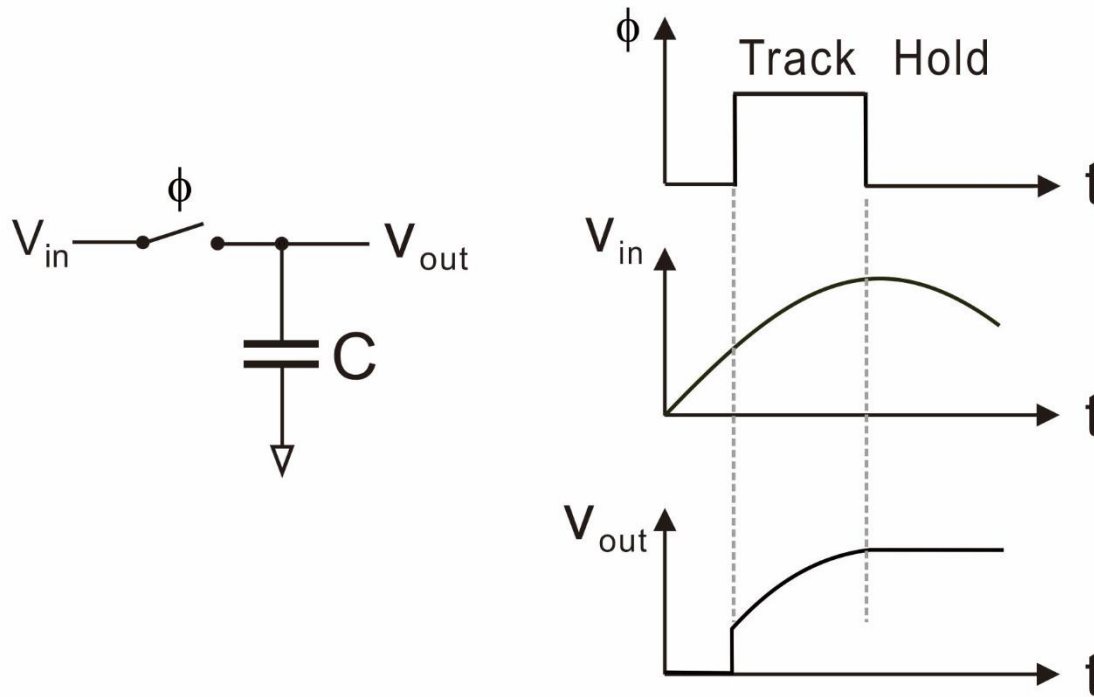
Nan Sun  
Tsinghua University  
nansun@Tsinghua.edu.cn

# ADC Big Picture



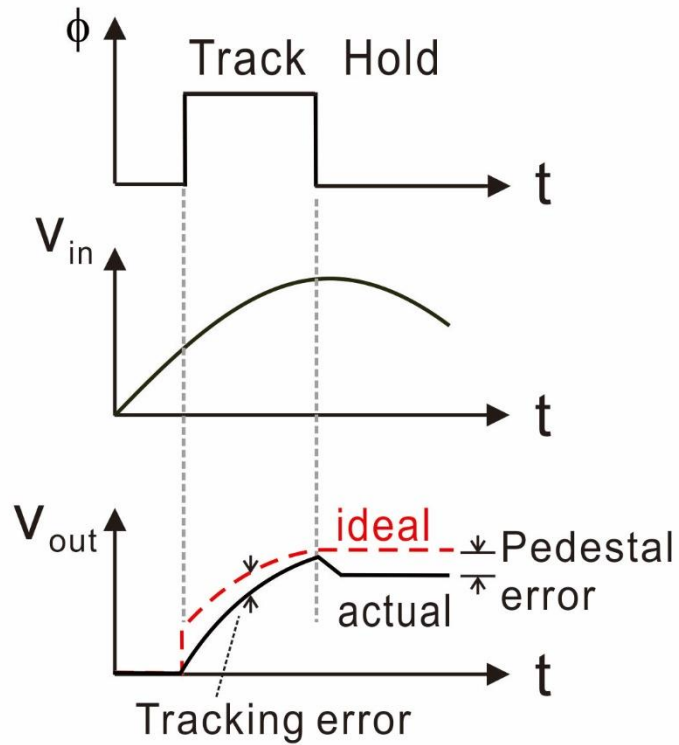
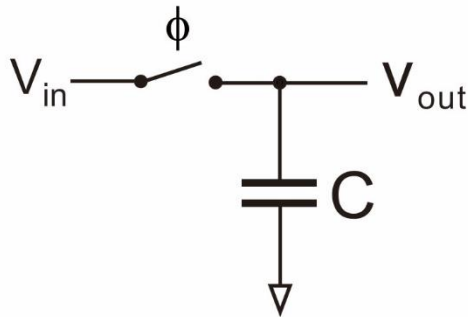
- How to build a sampling circuit in the real-world?
  - Ideal Dirac sampling is impractical
  - Build a track and hold circuit

# Ideal Track-and-Hold Circuit



- Infinitely fast and accurate tracking of  $V_{in}$
- No error in the hold phase

# Circuit with MOS Switch



- Tracking error
- Pedestal error

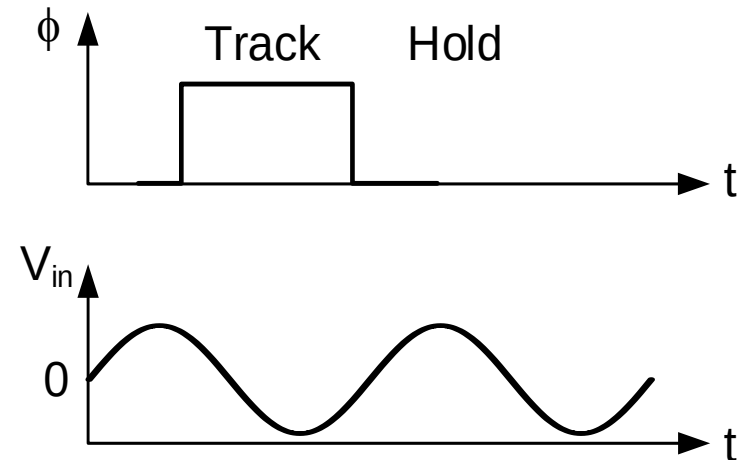
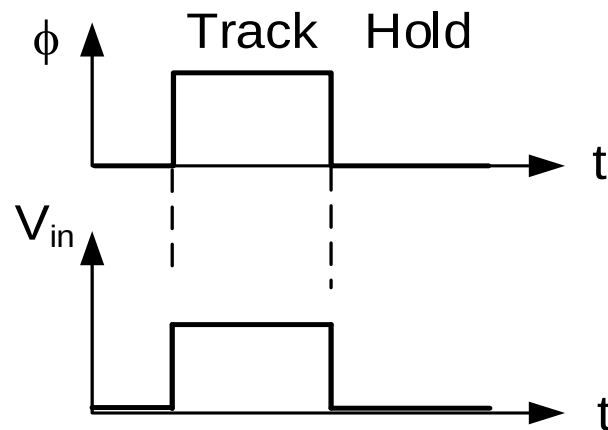
# Nonidealities

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- Switch resistance
  - Finite acquisition time
  - Thermal noise
  - Nonlinear resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities

# Finite Acquisition Time

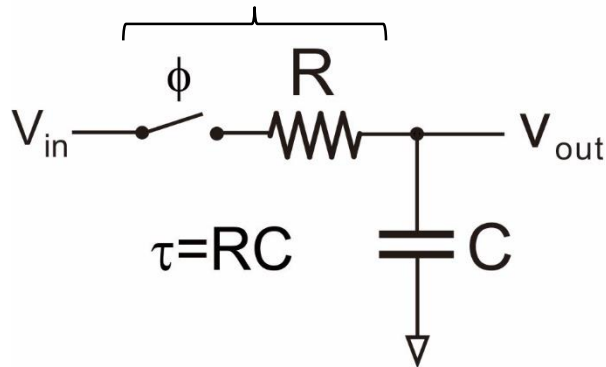
- Case 1: Input is a sampled data signal, e.g. the output of a switched capacitor circuit
- Case 2: Input is a varying continuous-time signal



# Finite Acquisition Time – Case 1

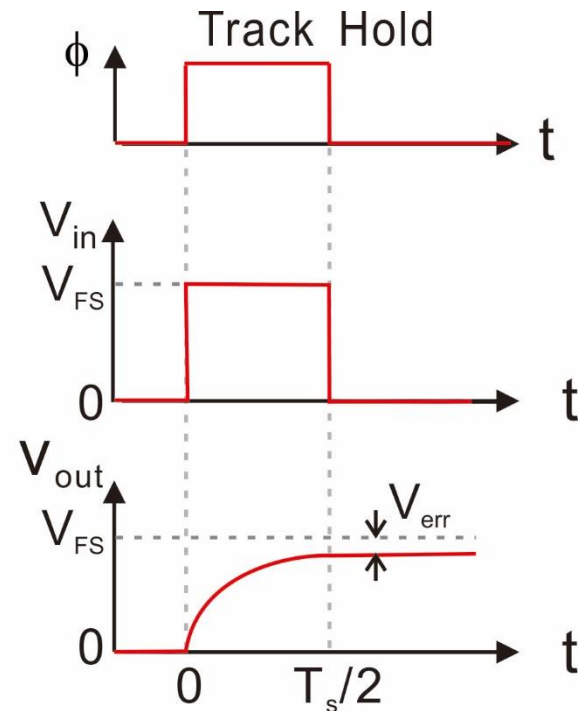
- For simplicity, let us ignore the finite rise time of  $V_{in}$
- Consider the worst case:  $V_{out}$  settles from 0 to  $V_{FS}$

1<sup>st</sup>-order MOS switch model



$$V_{out}(t) = V_{FS}(1 - e^{-t/\tau})$$

$$V_{err} = V_{out}(T_s/2) - V_{FS} = -V_{FS}e^{-T_s/2\tau}$$



# Finite Acquisition Time – Case 1

- We want the settling error  $|V_{\text{err}}| < \frac{1}{2} \text{ LSB}$ ?

$$V_{\text{FS}} e^{-T_s/2\tau} \leq \Delta/2 = V_{\text{FS}}/2^{B+1}$$

- The required # time constant N:

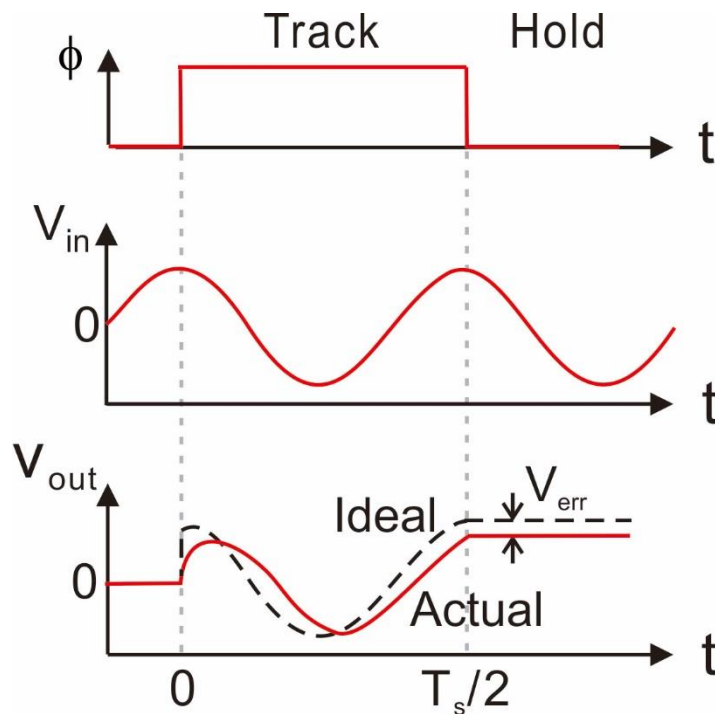
$$N \equiv \frac{T_s}{2\tau} > \ln(2^{B+1})$$

- Does settling error cause distortion?

| B  | N     |
|----|-------|
| 6  | >4.9  |
| 10 | >7.6  |
| 14 | >10.4 |
| 18 | >13.2 |

# Finite Acquisition Time – Case 2

- Consider a sinusoidal input with zero offset



$$V_{in}(t) = A \cos(\omega t)$$

$$V_{out}(t) \approx A e^{-t/\tau} + \frac{A}{\sqrt{1 + \omega^2 \tau^2}} \cos(\omega(t - \tau))$$

Annotations for the equation:

- Amplitude attenuation:** Points to the coefficient  $\frac{A}{\sqrt{1 + \omega^2 \tau^2}}$ .
- Delay:** Points to the phase shift  $\tau$  in the cosine argument.
- Initial discontinuity settling:** Points to the  $A e^{-t/\tau}$  term.
- Steady-state RC filter response:** Points to the  $\frac{A}{\sqrt{1 + \omega^2 \tau^2}} \cos(\omega(t - \tau))$  term.

# Finite Acquisition Time – Case 2

---

- The error at  $t=T_s/2$  consists of two parts:
  - Error due to the exponential decay of the initial discontinuity
    - Similar to Case 1, it decays with time
  - Error in the steady-state term
    - Result of RC LPF
      - Magnitude attenuation and phase shift
      - It depends on RC (filter response) and the input frequency
      - It does not decay with time
    - Will this error cause distortion?
    - How large is the amplitude attenuation?

# Finite Acquisition Time – Case 2

- Let us calculate the percent amplitude error

$$A_{\text{err}} = 1 - \frac{1}{\sqrt{1 + (\omega\tau)^2}} = 1 - \frac{1}{\sqrt{1 + \left(\frac{\pi f_{\text{in}}}{N f_s}\right)^2}} \approx \frac{1}{2} \left(\frac{\pi f_{\text{in}}}{N f_s}\right)^2$$

- Amplitude error increases with  $f_{\text{in}}$ , but decreases with  $N$ .

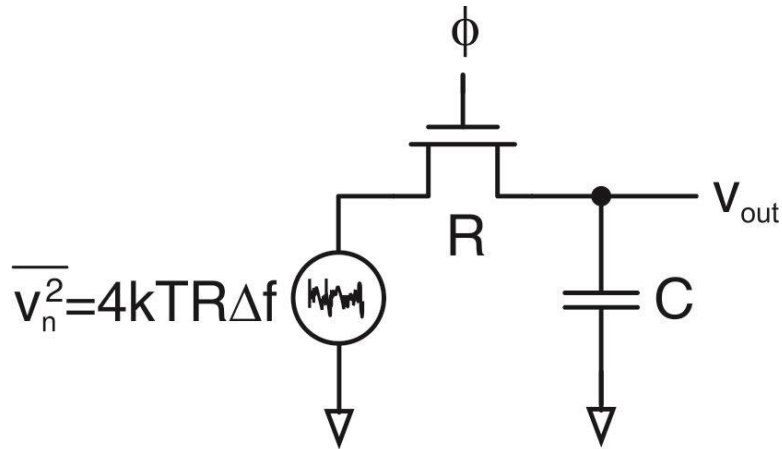
| B  | N    | $A_{\text{err}} (f = f_s/10)$ | $A_{\text{err}} (f_{\text{in}} = f_s/2)$ |
|----|------|-------------------------------|------------------------------------------|
| 6  | 4.9  | 0.20%                         | 5%                                       |
| 10 | 7.6  | 0.08%                         | 2%                                       |
| 14 | 10.4 | 0.05%                         | 1%                                       |
| 18 | 13.2 | 0.03%                         | 0.7%                                     |

# Summary – Finite Acquisition Time

---

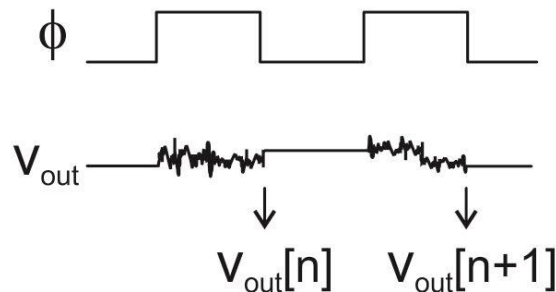
- For sampled-data input, we can achieve accurate tracking given a sufficient amount of time.
- For continuous input, we need to consider the errors due to both the initial settling and **the filter response**.
  - In applications where attenuation is intolerable, we may need to further reduce the RC time constant.
  - In applications where attenuation is tolerable, the RC time constant is usually set by the distortion specs.
    - **The larger the attenuation, the larger the instantaneous voltage drop across the nonlinear MOSFET → undesired harmonics.**
- What about the phase delay?
  - The sampled input is always a delayed ( $\tau$ ) version.

# Thermal Noise



- Transistor thermal noise dominates
- $1/f$  noise can be ignored
  - $1/f$  noise is like an offset

$$\overline{\frac{v_{out}^2}{\Delta f}} = 4kTR \left| \frac{1}{1 + sRC} \right|^2$$



$$\overline{|v_{out}^2[n]|} = 4kTR \int_0^\infty \frac{1}{1 + (2\pi fRC)^2} df = \frac{kT}{C}$$

# Equipartition Theorem

---

- Equipartition theorem in statistical physics states that in thermal equilibrium each degree of freedom holds on average thermal energy of  $kT/2$ .
- In an electronic circuit, each independent capacitor holds an electric energy of  $kT/2$  and each independent inductor holds a magnetic energy of  $kT/2$ .
- This theorem can help reduce the complexity of noise analysis.
- Use this theorem cautiously!

$$\begin{aligned}\frac{1}{2}C\overline{v_{out}^2} &= \frac{1}{2}kT \\ \Rightarrow \overline{v_{out}^2} &= \frac{kT}{C}\end{aligned}$$

# kT/C Noise

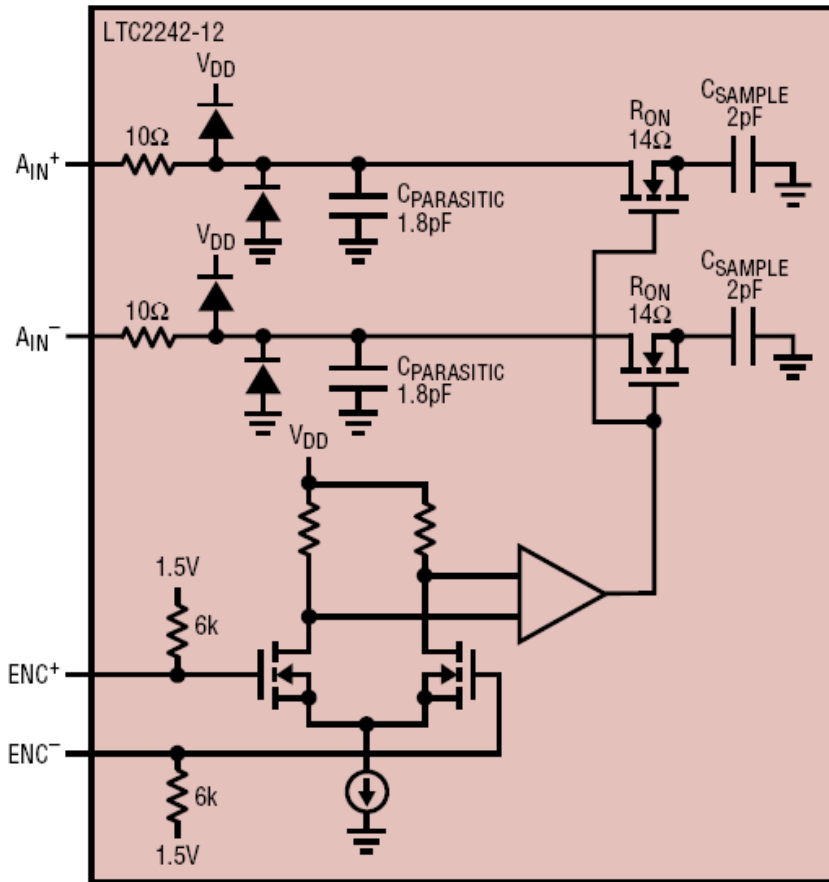
- Assuming that we make the kT/C noise equal to the quantization noise of a B-bit ADC

$$\frac{kT}{C} = \frac{\Delta^2}{12} \Rightarrow C = 12kT \cdot \left(\frac{2^B}{V_{FS}}\right)^2$$

- Assuming  $V_{FS} = 1V$ ,

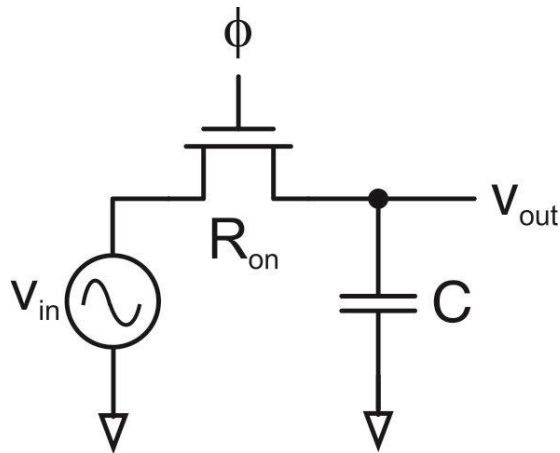
| B  | C [pF]       |
|----|--------------|
| 8  | <b>0.003</b> |
| 10 | 0.052        |
| 12 | 0.834        |
| 14 | 13.3         |
| 16 | 213          |
| 18 | <b>3416</b>  |

# Real-World Example



- LTC 2242
- 12-bit ADC
- Why 2pF?

# Voltage Dependence of $R_{on}$



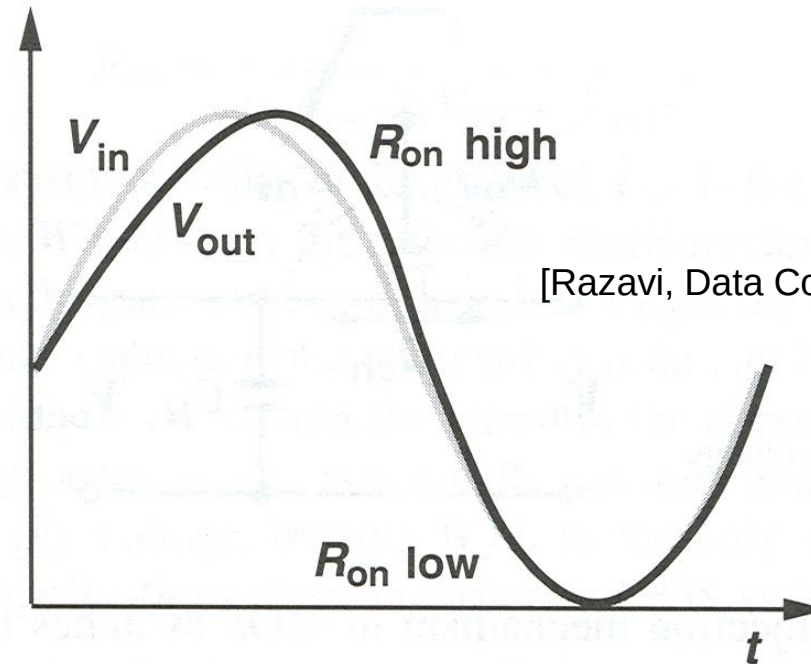
$$I_D = \mu C_{ox} \frac{W}{L} \left( V_{GS} - V_{th} - \frac{V_{DS}}{2} \right) V_{DS}$$

$$R_{ON} \approx \left[ \frac{\partial I_D}{\partial V_{DS}} \Big|_{V_{DS} \rightarrow 0} \right]^{-1} = \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

$$R_{ON} \approx \frac{1}{\mu C_{ox} \frac{W}{L} (\phi - V_{IN} - V_{th})}$$

- $R_{ON}$  is dependent on  $V_{in}$

# Track Mode Nonlinearity



[Razavi, Data Conversion System Design, p.16]

- $V_{out}$  tracks  $V_{in}$  well for small  $V_{in}$ 
  - Large  $V_{in} \rightarrow$  large  $R_{on} \rightarrow$  distortion

# Analysis

---

$$I_D \approx K(V_{GS} - V_{th})V_{DS} - \frac{K}{2}V_{DS}^2$$

$$C \frac{dV_{out}}{dt} = I_D \approx K(\varphi - V_{out} - V_{th})(V_{in} - V_{out}) - \frac{K}{2}(V_{in} - V_{out})^2$$

- We can solve the above equation use Volterra Series analysis
  - General method that allows us to calculate the frequency domain response of nonlinear circuits with memory
  - See [Yu, TCAS II, 2/1999]

# Result

---

$$HD_3 = \frac{\text{Amplitude of the third harmonic}}{\text{Amplitude of the fundamental}} \cong \frac{1}{4} \frac{A^2}{(V_{GS} - V_{th})^2} \cdot \omega\tau$$

- $V_{GS}$  is the value of the gate-source voltage when AC input is zero.
- For low distortion
  - Make amplitude  $A$  smaller than  $(V_{GS} - V_{th})$ 
    - Low swing  $\rightarrow$  SNR decreases
  - Make  $\omega$  smaller
    - Low frequency  $\rightarrow$  speed decreases
  - Make  $\tau = RC$  smaller
    - Big switch  $\rightarrow$  parasitic caps increases and driving power goes up

# Example

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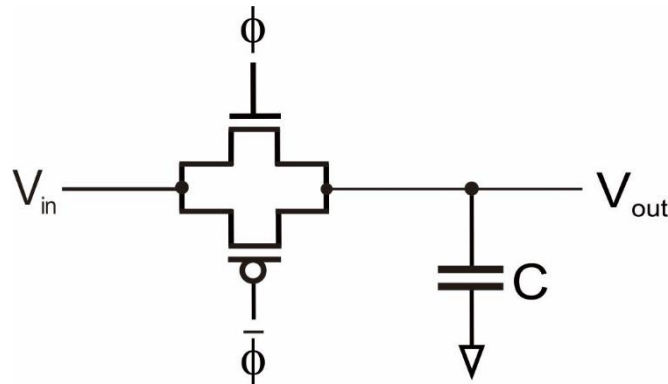
- Parameters
  - $V_{DD} = V_{CLK} = 1.8V$
  - $V_{in}$  is centered at  $V_{DD}/2 = 0.9V$
  - $V_{GS} - V_{th} = 1.8V - 0.9V - 0.45V = 0.45V$
  - $A = 0.4V$
  - $N = 0.5T_s/\tau = 10$
  - $f_{in} = f_s/2$

$$HD_3 \approx \frac{\pi}{4} \frac{0.4^2}{(0.45)^2} \cdot \left( \frac{1}{20} \right) = -30 \text{ dB}$$

- The nonlinearity is huge! How can we reduce it?

# CMOS Switch

- Adding a PMOS can reduce the resistance variation
  - As  $V_{IN}$  increases, NMOS resistance goes up, but PMOS resistance goes down



$$R \approx \frac{1}{\mu_n C_{ox} \left[ \frac{W}{L} \right]_n (V_{GSn} - V_{thn})} \parallel \frac{1}{\mu_p C_{ox} \left[ \frac{W}{L} \right]_p (|V_{GSp}| - |V_{thp}|)}$$

# Analysis

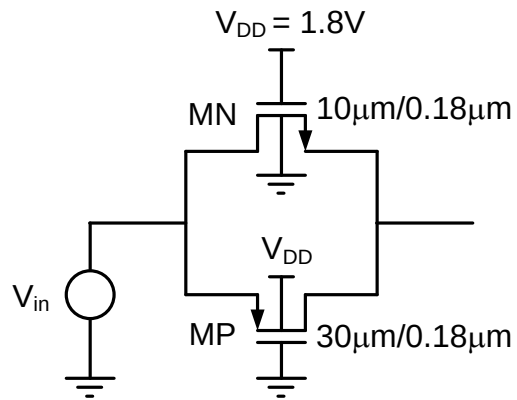
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$$R \approx \frac{1}{\mu_n C_{ox} \left[ \frac{W}{L} \right]_n (V_{DD} - V_{thn}) - \left( \mu_n C_{ox} \left[ \frac{W}{L} \right]_n - \mu_p C_{ox} \left[ \frac{W}{L} \right]_p \right) V_{in} - \mu_p C_{ox} \left[ \frac{W}{L} \right]_p |V_{thp}|}$$
$$\approx \frac{1}{\mu_n C_{ox} \left[ \frac{W}{L} \right]_n (V_{DD} - V_{thn} - |V_{thp}|)}$$

if  $\mu_n \left[ \frac{W}{L} \right]_n = \mu_p \left[ \frac{W}{L} \right]_p$

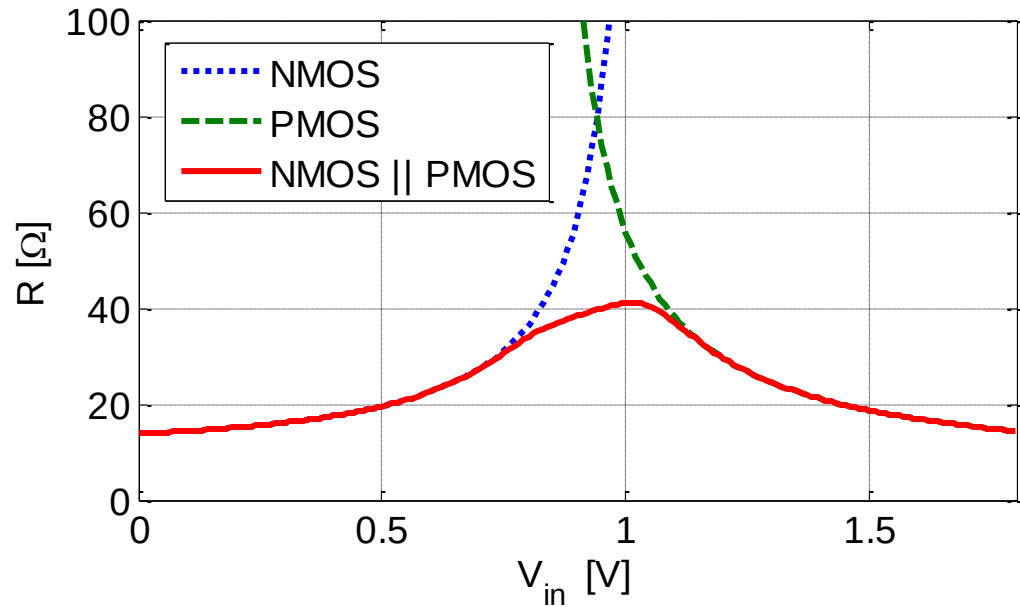
- R is independent from  $V_{in}$ ; however, this model is over-simplified.
- Missing factors
  - Short channel effects
  - Back-gate effect

# Real CMOS Switch



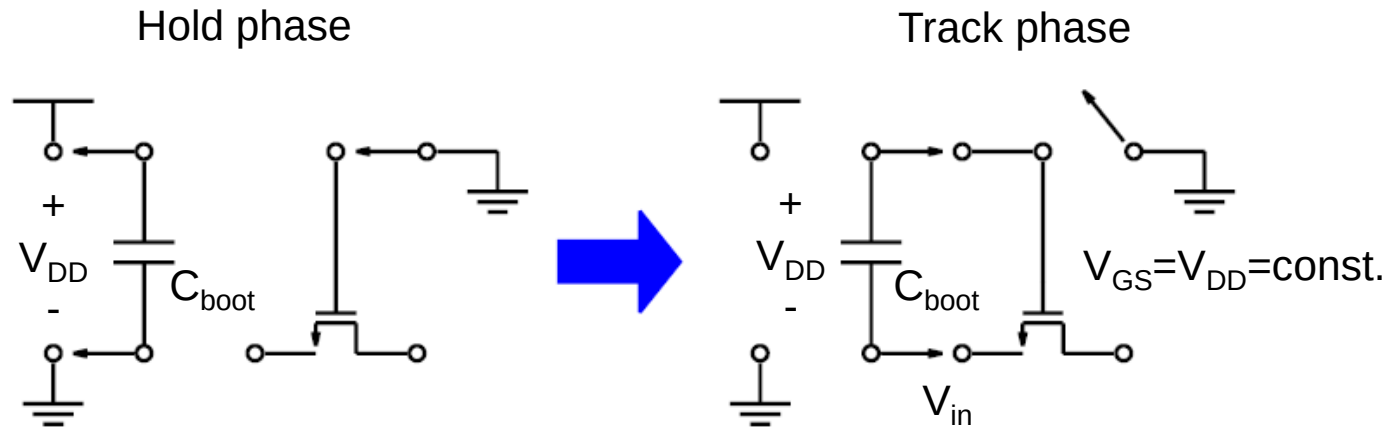
From B. Murmann, EE 315, Ch.

5



- There are still large resistance variations. The benefit is limited.

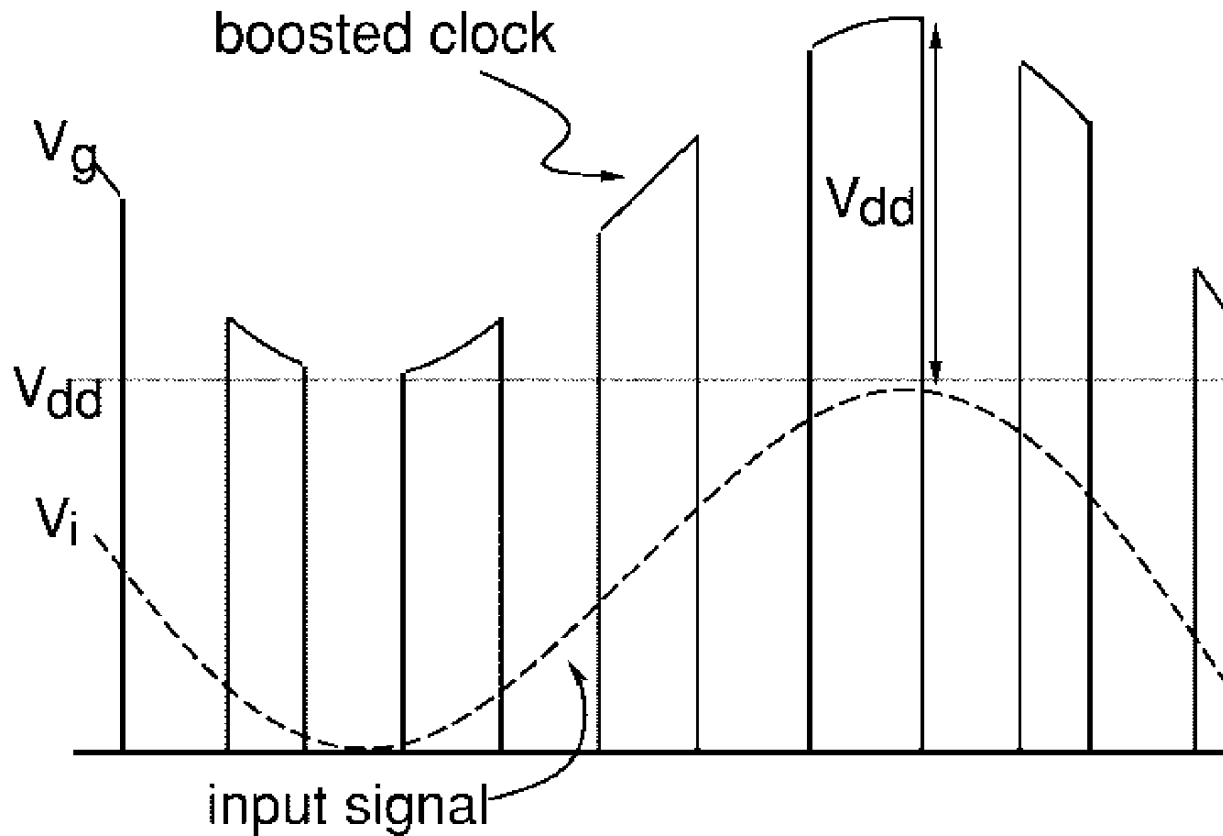
# Clock Bootstrapping



A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999.

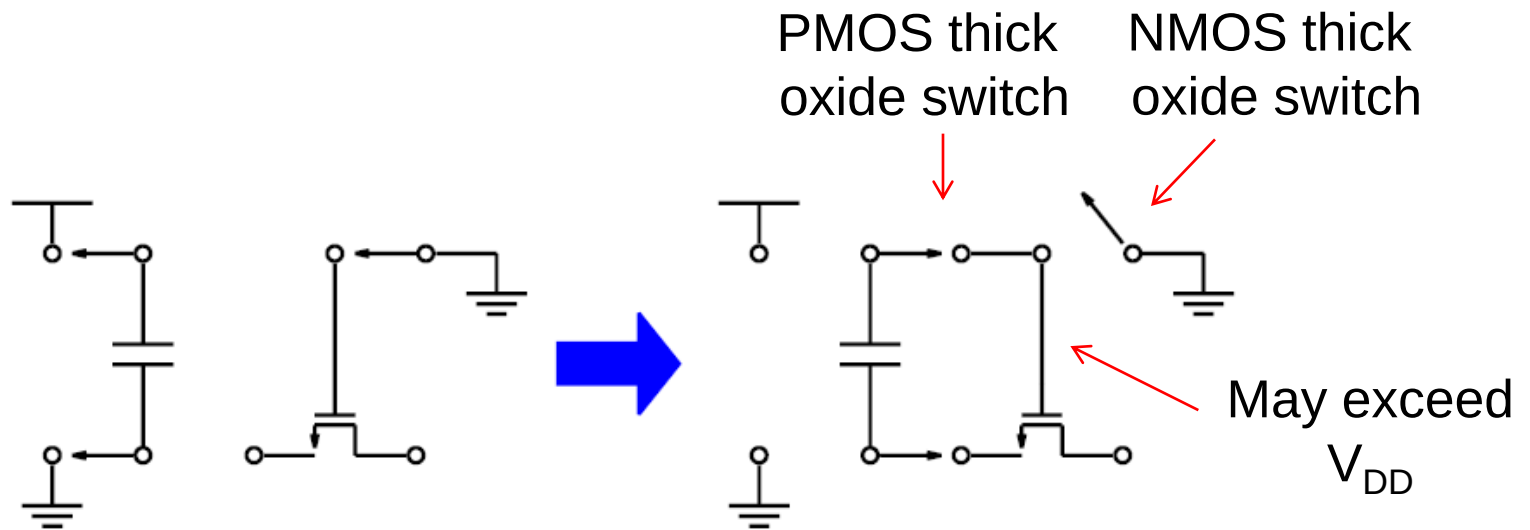
- Hold phase
  - Sampling switch is off
  - $C_{boot}$  is precharged to  $V_{DD}$
- Track phase
  - Sampling switch is on with  $V_{GS} = V_{DD} = \text{const}$
  - Thus,  $R_{on}$  is signal independent

# Waveforms



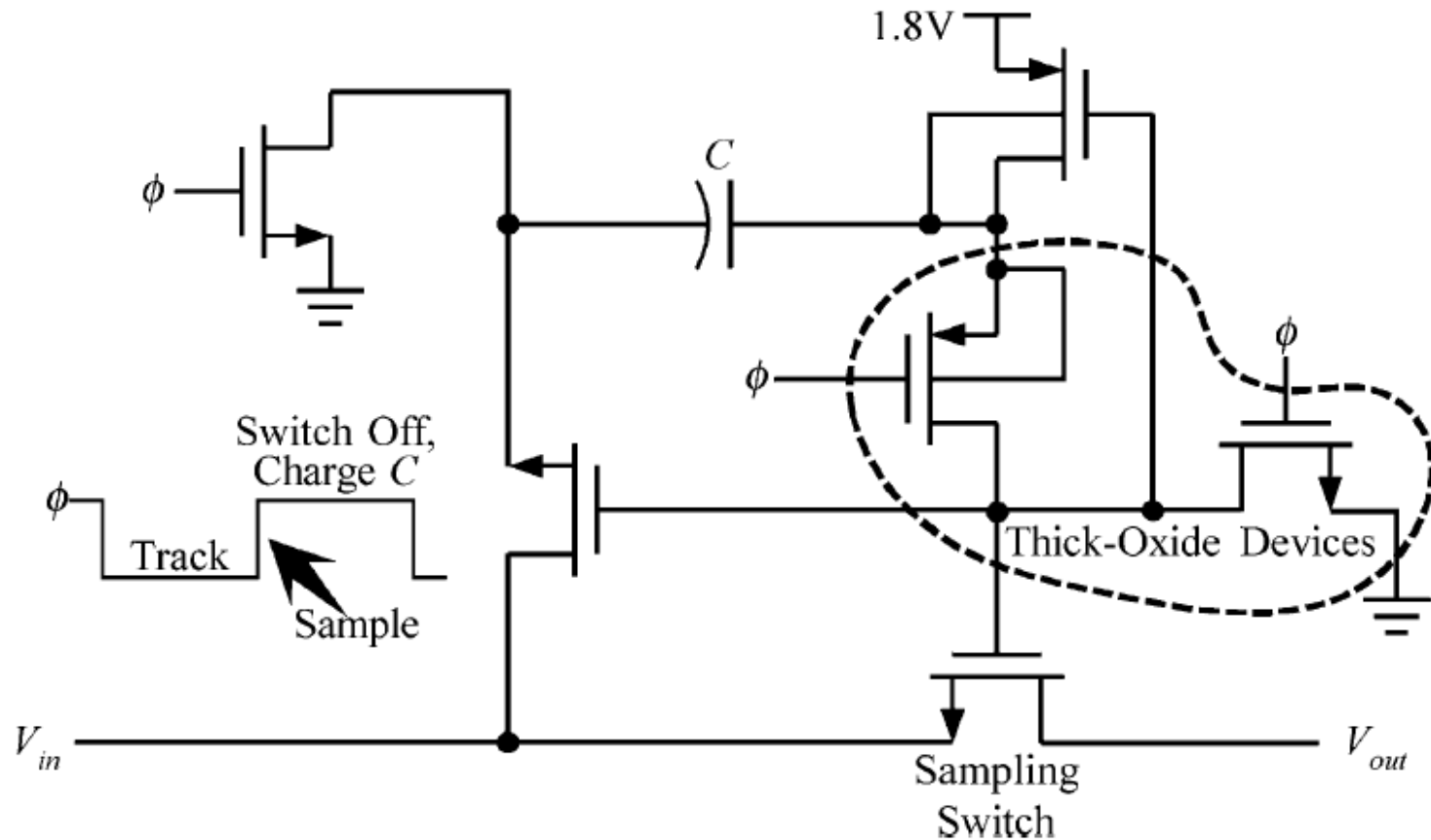
A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999.

# Implementation Issues



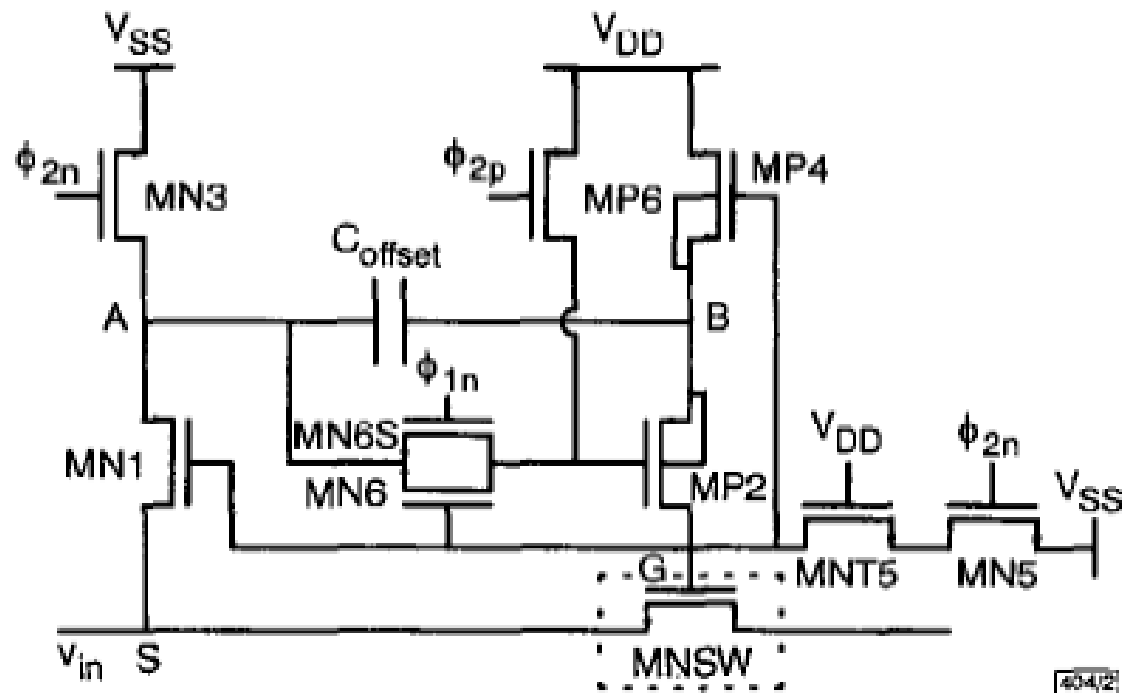
- $V_{GS}$  or  $V_{GD}$  may exceed  $V_{DD}$ , causing transistor breakdown.
- Thick oxide devices can be used to ensure reliability, leading to increased cost, larger parasitic capacitance, and larger resistance.

# With Thick-Oxide Devices



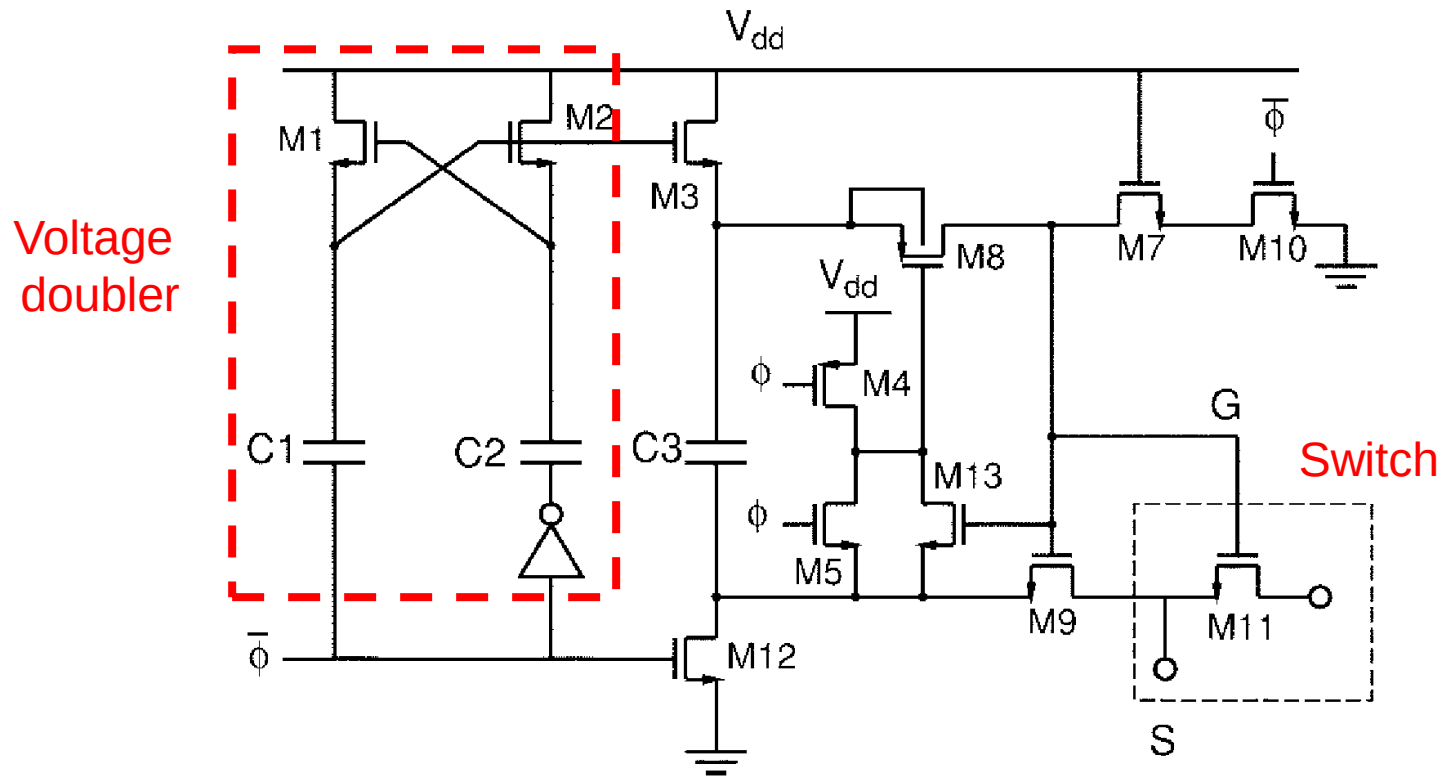
Siragusa and Galton, "A digitally enhanced 1.8-V 15-bit 40-Msample/s CMOS pipelined ADC," JSSC, Dec. 2004.

# Without Thick-Oxide Devices



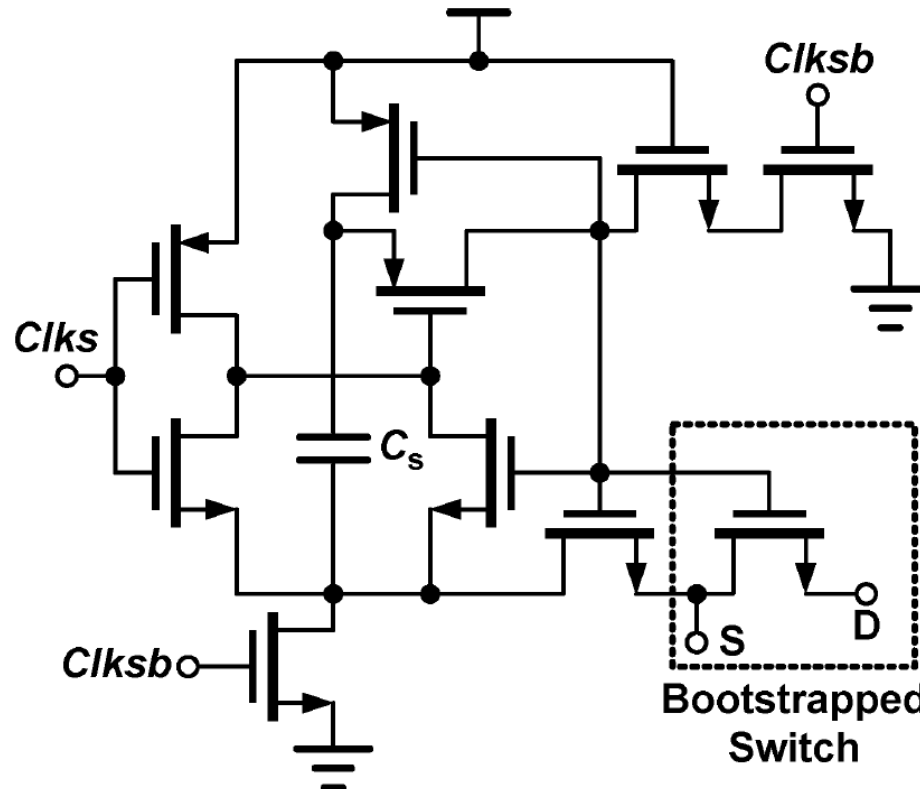
Dessouky and Kaiser, "Input switch configuration suitable for rail-to-rail operation of switched opamp circuits," Electronics Letters, Jan. 1999.

# Another Implementation



A. Abo et al., "A 1.5-V, 10-bit, 14.3-MS/s CMOS Pipeline Analog-to-Digital Converter," IEEE J. Solid-State Circuits, pp. 599, May 1999

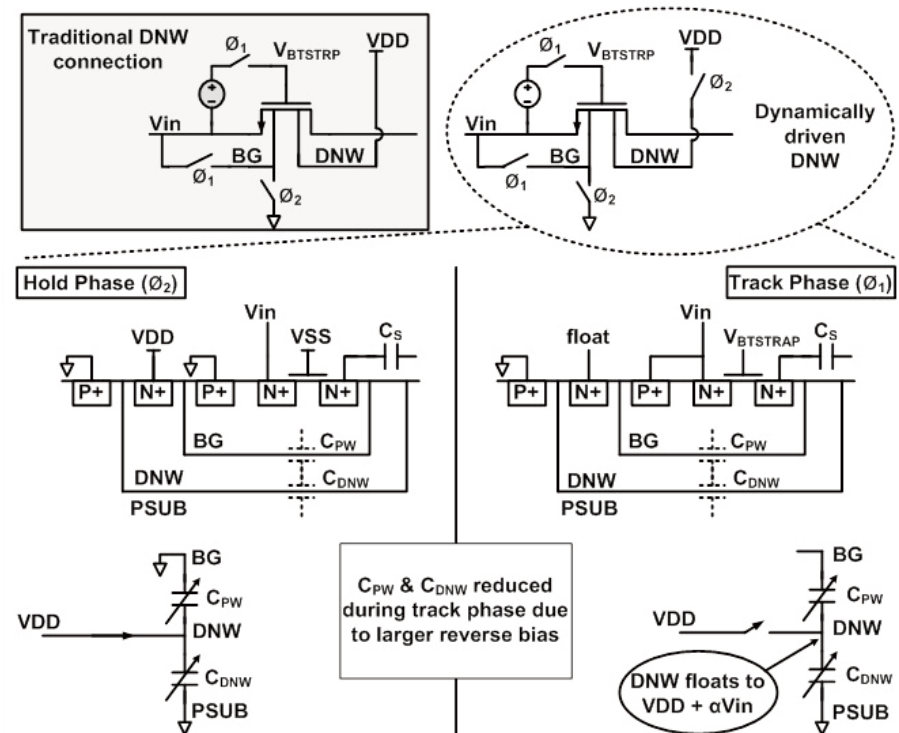
## Simplified Implementation



C.-C. Liu, "A 10-bit 50-MS/s SAR ADC with a monotonic capacitor switching procedure," IEEE J. Solid-State Circuits, pp. 731, 2010

# Limitation of Bootstrap Technique

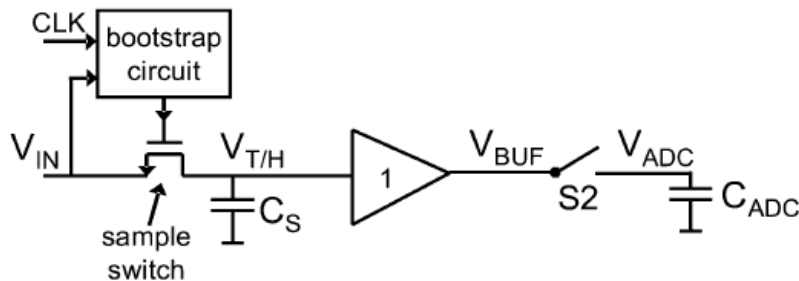
- Parasitic capacitance at the gate of the sampling switch
  - Capacitive divider  $\rightarrow V_{GS}$  not a constant
  - Can be alleviated by increasing the size of the bootstrap capacitor
    - However, limited by the intrinsic parasitic cap associated with the bootstrap cap
- Back-gate effect
  - Can be alleviated by using triple well process



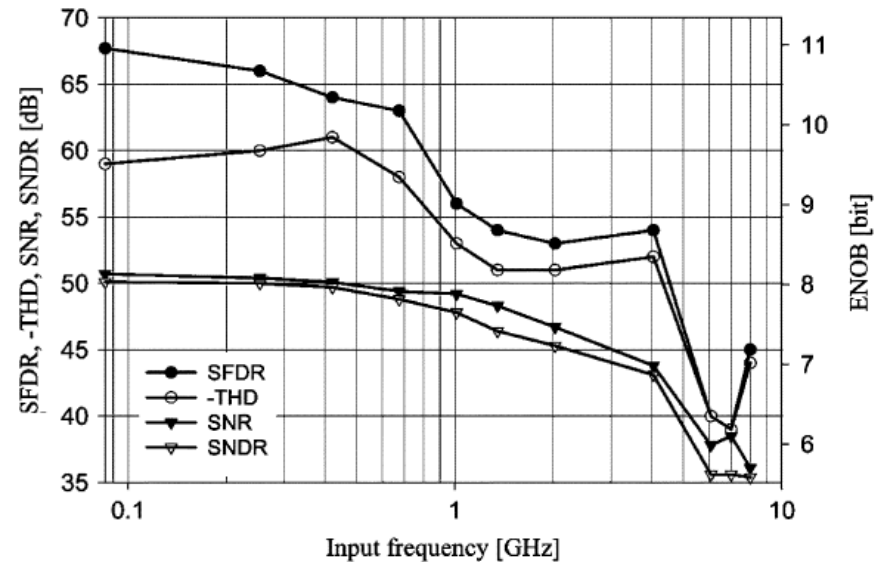
Brunsilius, ISSCC 2011, 10.4, pp. 186

# Performance of Bootstrapped Sampler

- Bootstrapped sampling tends to work well up to 60 dB linearity
  - Now the linearity bottleneck is related to the switch capacitance.
- Example



[Louwsma, JSSC 4/2008]

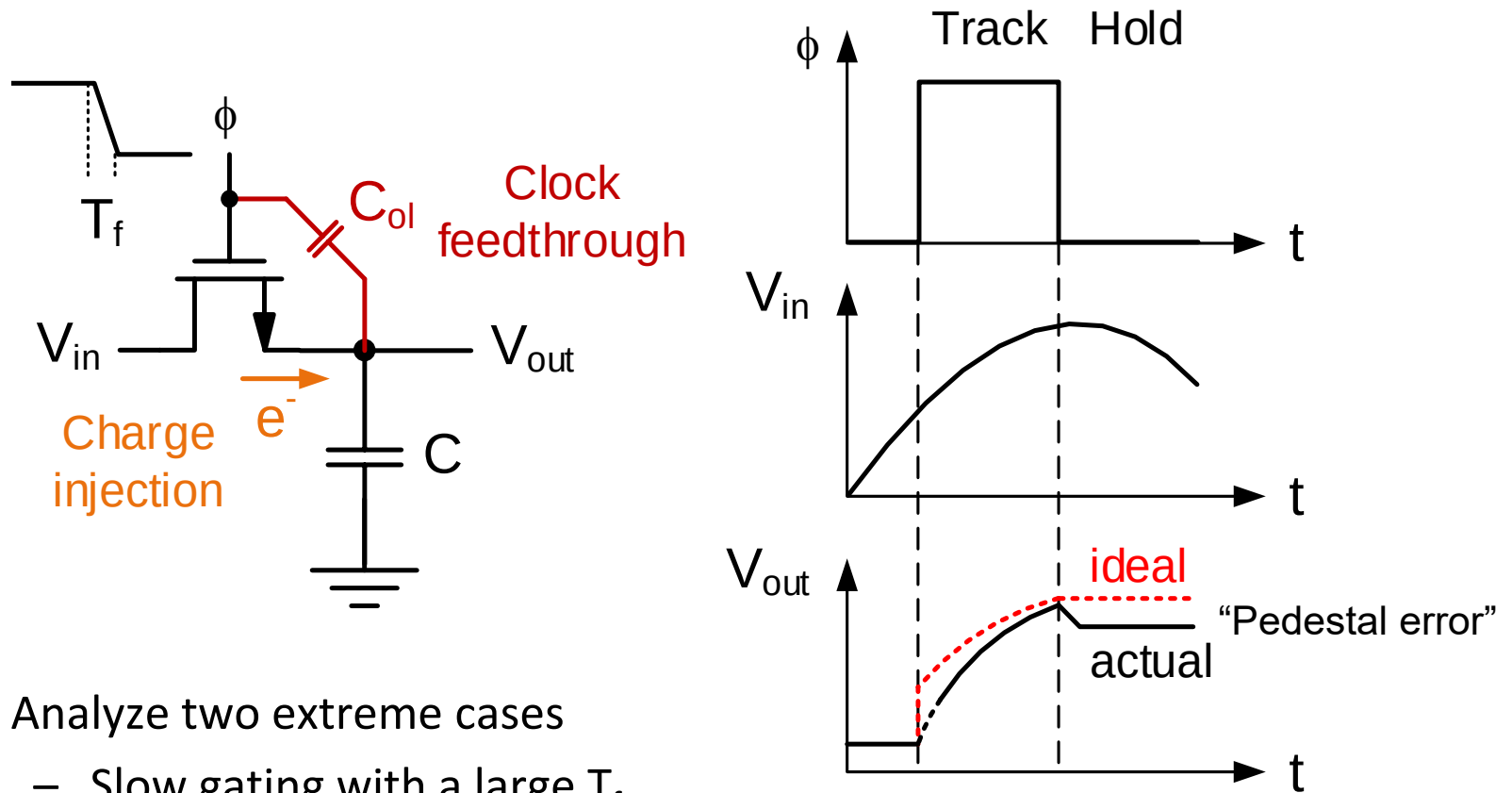


# Nonidealities

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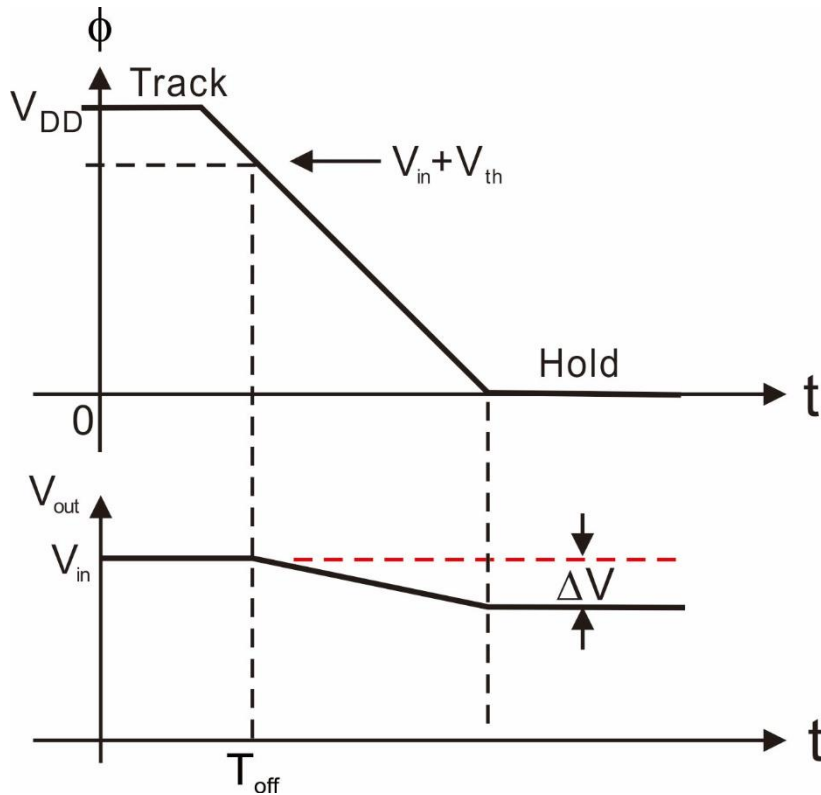
- Switch resistance
- Switch capacitance
  - Clock feedthrough
  - Channel charge injection
- Sampling clock jitter
- Other nonidealities

# Charge Injection & Clock Feedthrough



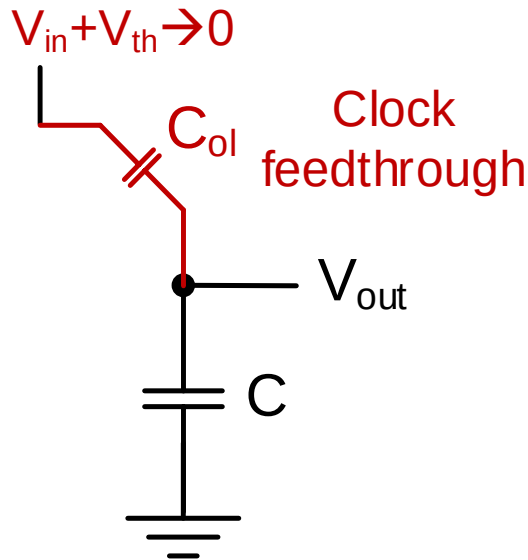
- Analyze two extreme cases
  - Slow gating with a large  $T_f$
  - Fast gating with a small  $T_f$

# Slow Gating



- All channel charges have disappeared by  $T_f$ . They are absorbed by the input source, and thus, do not introduce error.
- Pedestal error is only due to clock feedthrough.

# Slow Gating Model for $t > T_f$



$$V_{out} = V_{in} - \Delta V = V_{in} - \frac{C_{ol}}{C_{ol} + C} (V_{in} + V_{th})$$

$$= V_{in}(1 + \varepsilon) + V_{os}$$

switch open

$$\varepsilon = -\frac{C_{ol}}{C_{ol} + C}$$

Gain Error

$$V_{os} = -\frac{C_{ol}}{C_{ol} + C} V_{th}$$

Offset Error

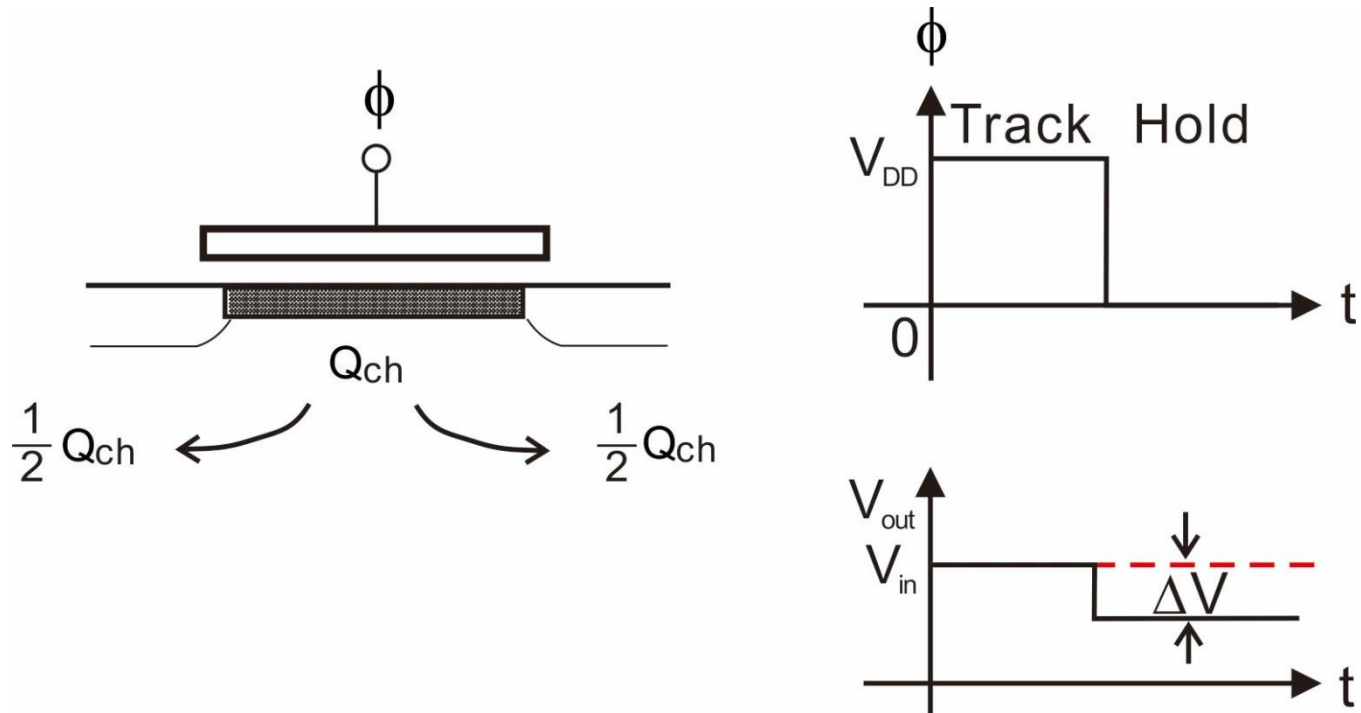
- Example:  $C=1\text{pF}$ ,  $C_{ol}=2\text{fF}$ , and  $V_{th}=0.45\text{V}$

$$\varepsilon = -0.2\%$$

$$V_{os} \approx -1\text{mV}$$

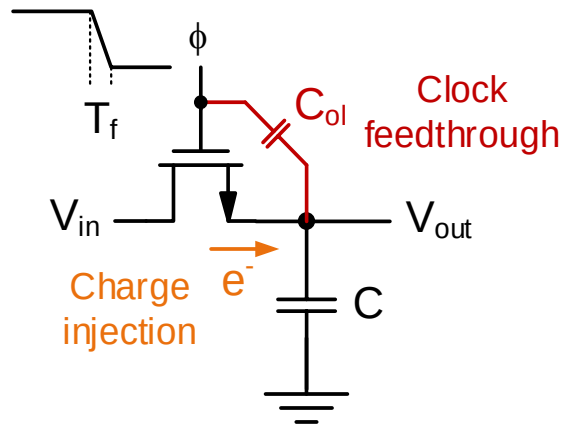
- Does this cause nonlinearity?

# Fast Gating



- Channel charge  $Q_{ch}$  has to flow to the input and the output nodes
- Under fast gating, we assume a 50/50 split

# Fast Gating Model for $t > T_f$



$$V_{out} = V_{in} - \frac{C_{ol}}{C_{ol} + C} V_{DD} + \frac{Q_{ch}}{2C} = V_{in}(1 + \epsilon) + V_{os}$$

$$\text{where } Q_{ch} = C_{ox}WL(V_{DD} - V_{in} - V_{th})$$

$$\epsilon = \frac{1}{2} \frac{C_{ox}WL}{C}$$

$$V_{OS} = -\frac{C_{ol}}{C_{ol} + C} V_{DD} + \frac{1}{2} \frac{C_{ox}WL}{C} (V_{DD} - V_{th})$$

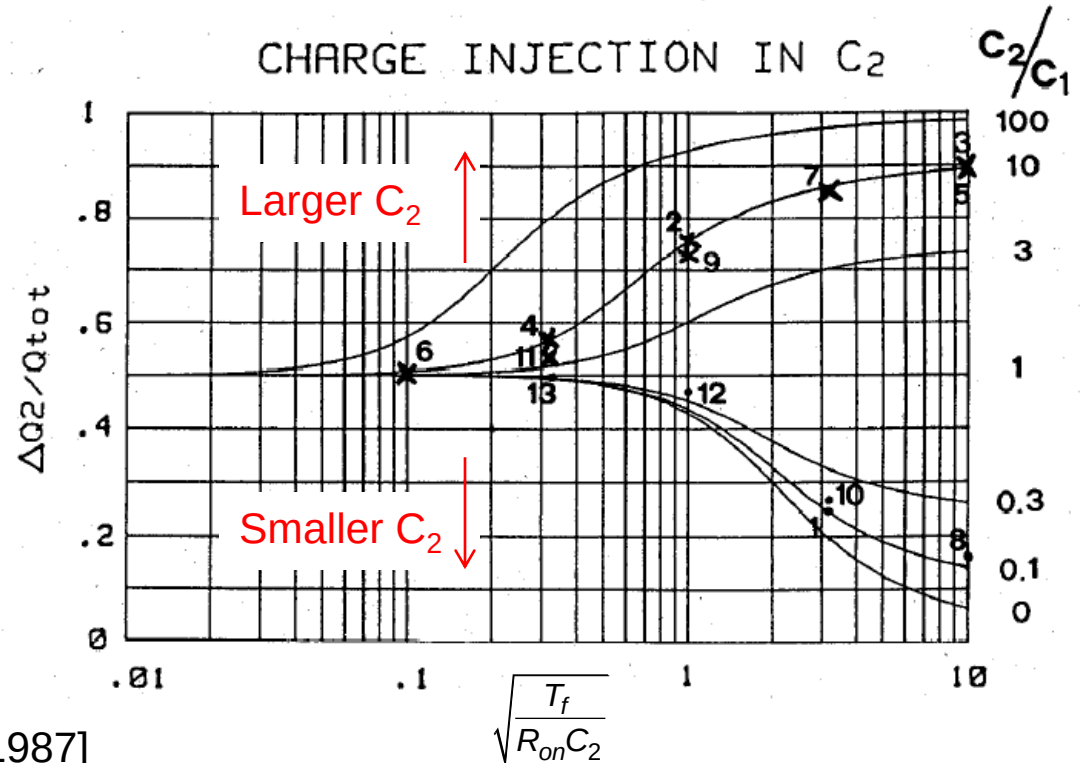
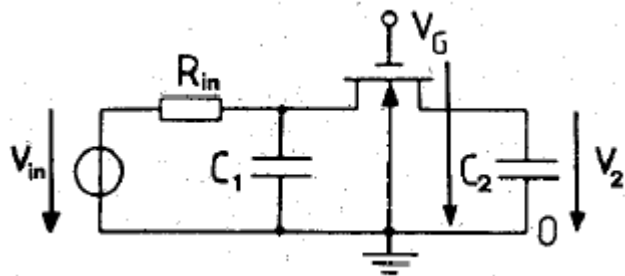
- Example:  $C=1\text{pF}$ ,  $V_{DD}=1.8\text{V}$ ,  $V_{th}=0.45\text{V}$ ,  $W=20\mu\text{m}$ ,  $LC_{ox}=2\text{fF}/\mu\text{m}$ , and  $C_{ol}=2\text{fF}$

$$\epsilon = 2\%$$

$$V_{OS} \approx -31\text{mV}$$

- Does this cause nonlinearity?

# Charge Split Ratio Data



[G. Wegmann et al., IEEE JSSC, 1987]

- In practice,  $R_{on}C_2$  and  $T_f$  are usually comparable
- Charge split depends on the impedances on both sides
  - More charge will go to the side with lower impedance

# Summary

---

- In practice, the situation is in between slow gating and fast gating.
- Slow gating shows less gain and offset error
  - Unsuitable for high-speed
  - More later
- Are the gain error and the offset error really constant?
  - No, our first-order model is not accurate enough.
  - We have ignored back-gate effect and short-channel effect
  - In practice, about 20% or 30% of what we calculated are nonlinear errors

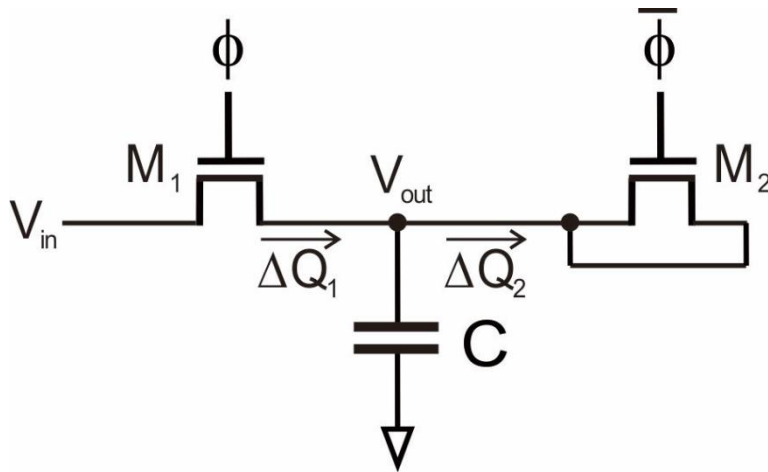
# Does CMOS Scaling Help?

---

$$\left. \begin{aligned} \Delta V &\approx \frac{1}{2} \frac{Q_{\text{ch}}}{C} \\ R &\approx \frac{1}{\mu C_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}})} = \frac{L^2}{\mu Q_{\text{ch}}} \end{aligned} \right\} \Delta V \cdot R \approx \frac{L^2}{2\mu C}$$

- For the same R (speed), channel-charge induced error  $\Delta V$  decreases as technology advances
  - Smaller L and larger  $\mu$
- For the same  $\Delta V$ , R is smaller, leading to faster speed.
- In general, switches benefit from CMOS scaling!

# Charge Cancellation



$$\Delta Q_1 \approx \frac{1}{2} Q_{ch1} + Q_{ov1}$$

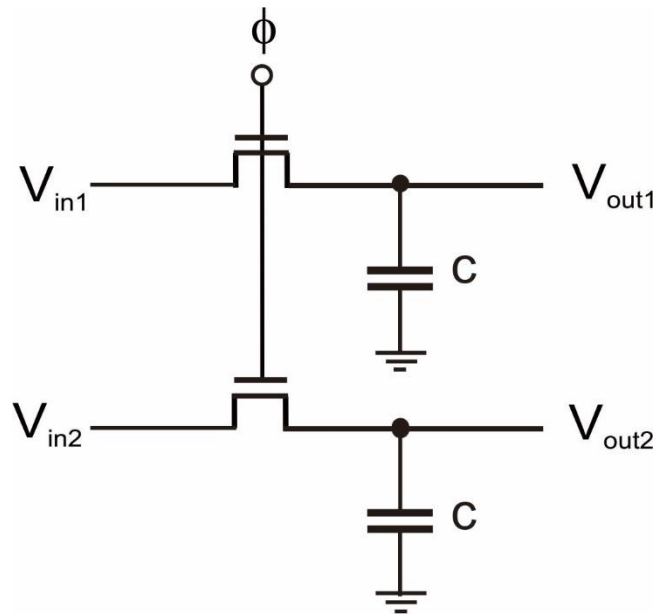
$$\Delta Q_2 \approx Q_{ch2} + 2Q_{ov2}$$

$$\Rightarrow W_2 = W_1/2, \quad L_2 = L_1$$

[Eichenberger and Guggenbühl, JSSC 8/89]

- Perfect cancellation relies on
  - Exactly 50% channel charge splitting
- Limited improvement

# Differential Sampling

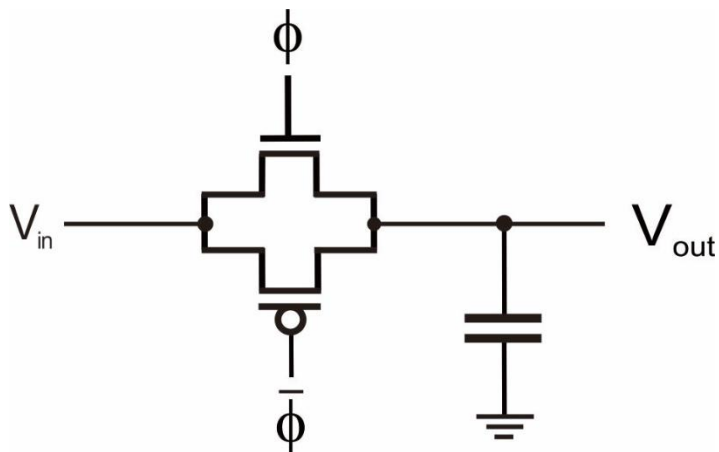


$$\begin{cases} V_{out1} = (1 + \varepsilon)V_{in1} + V_{OS} \\ V_{out2} = (1 + \varepsilon)V_{in2} + V_{OS} \end{cases}$$

$$\Rightarrow V_{out} = (1 + \varepsilon)V_{in}$$

- Offset will be cancelled, but the gain error remains the same.
  - Differential circuit only cancels out the common mode.

# CMOS Switch



$$Q_{\text{chn}} \approx -W_n L_n C_{\text{ox}} (V_{\text{DD}} - V_{\text{in}} - V_{\text{thn}})$$

$$Q_{\text{chp}} \approx W_p L_p C_{\text{ox}} (V_{\text{in}} - |V_{\text{thp}}|)$$

$$\Delta V_0 \approx \frac{Q_{\text{chn}} + Q_{\text{chp}}}{2C}$$

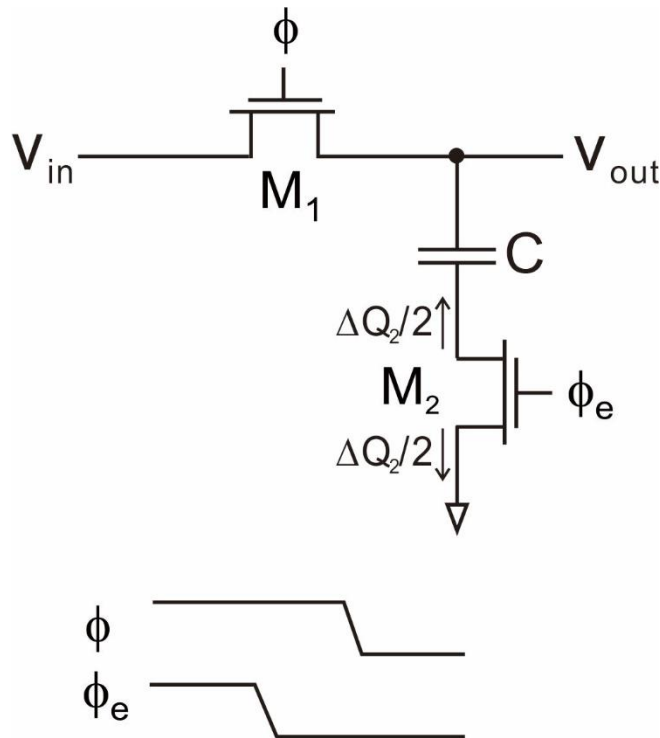
- Unfortunately, the signal dependent parts do not cancel, but add!

# Bottom Plate Sampling

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- Key idea: sample  $V_{in}$  at the "grounded" side of the capacitor to achieve signal independent sampling
- This technique allows us to achieve high linearity in combination with bootstrap technique
- Original papers:
  - D. J. Allstot and W. C. Black, "Technological Design Considerations for Monolithic MOS Switched-Capacitor Filtering Systems," Proc. IEEE, pp. 967-986, Aug. 1983.
  - K.-L. Lee and R. G. Meyer, "Low-Distortion Switched-Capacitor Filter Design Techniques," IEEE J. Solid-State Circuits, pp. 1103-1113, Dec. 1985.

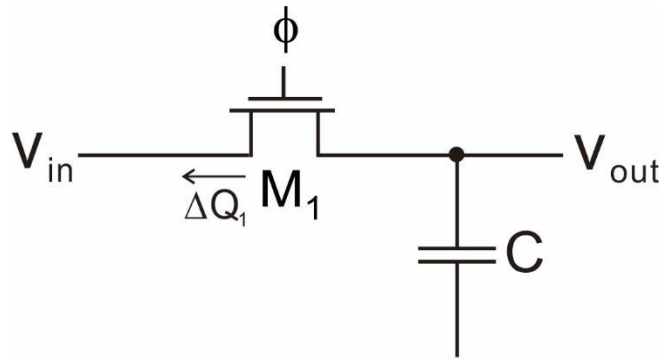
# Bottom Plate Sampling Analysis (1)



- Turn  $M_2$  off "slightly" before  $M_1$ 
  - Typically 100 ps difference
- When  $M_2$  is off, it injects charge
 
$$\Delta Q_2 \approx -WLC_{ox}(V_{DD} - V_{th})$$
- To 1st-order, the charge injected by  $M_2$  is signal independent
  - Assuming  $R_1$  does not depend on  $V_{in}$
- Voltage across  $C$  (50% split)
 
$$V_C = V_{in} - \frac{\Delta Q_2}{2C}$$

# Bottom Plate Sampling Analysis (2)

- The channel charge of  $M_1$  is signal dependent:

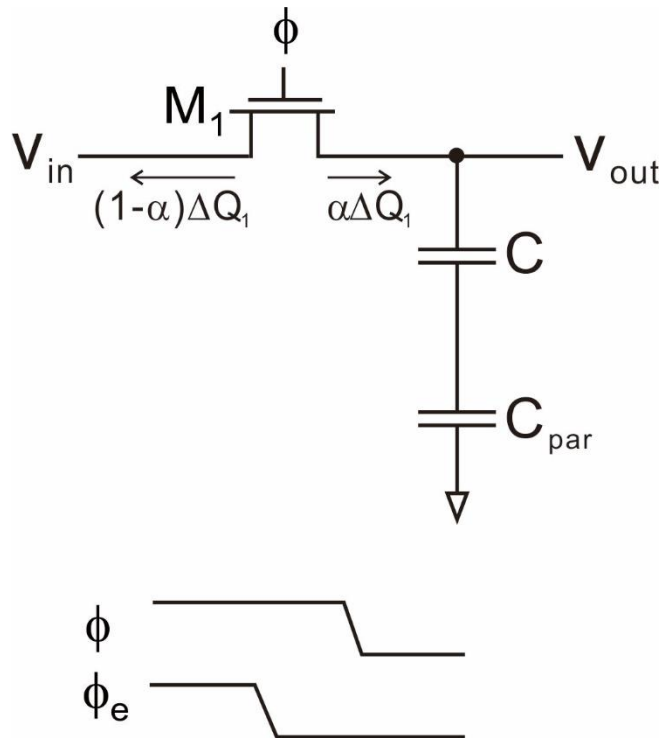


$$\Delta Q_1 \approx -WLC_{ox}(\phi - V_{in} - V_{th})$$

- However, when  $M_1$  is off, it cannot inject charge to  $C$  because its bottom side is open.
- This model is too ideal – we need to consider the parasitic cap

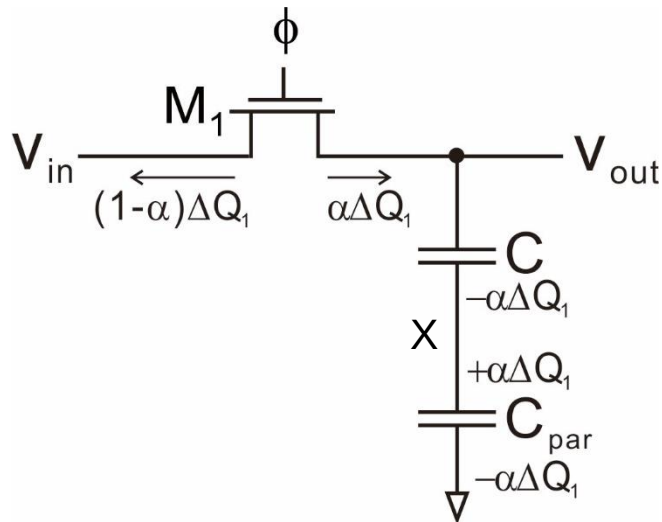


# Bottom Plate Sampling Analysis (3)



- There are parasitic capacitors
- $M_1$  will inject signal dependent charge onto the series combination of  $C$  and the parasitic capacitance at its bottom plate  $C_{par}$
- This seems that the improvement is quite limited, as  $V_{out}$  is affected.
- Nevertheless, if we consider the sampled charge instead of the voltage...

# Bottom Plate Sampling Analysis (4)



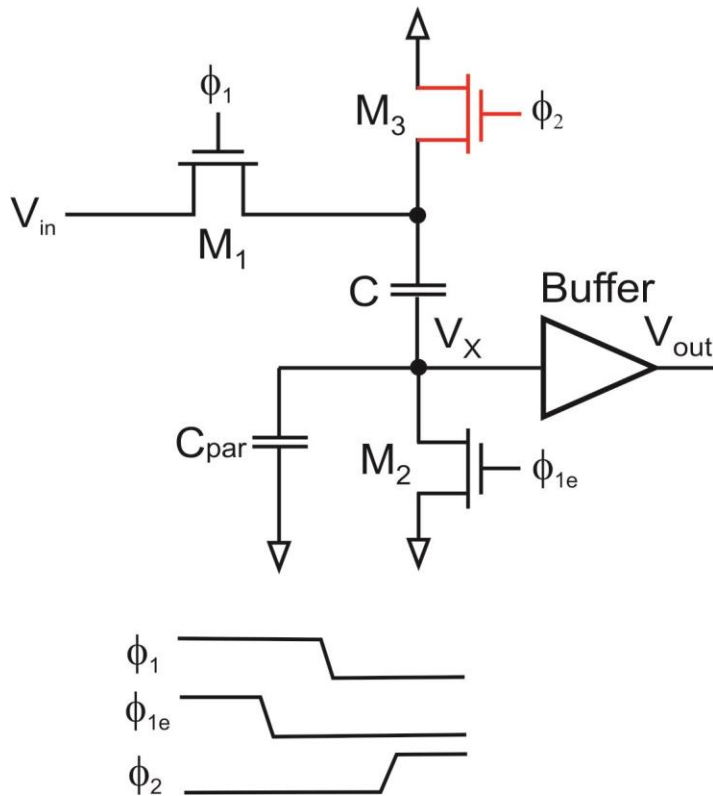
- Key observation:
  - Even though  $M_1$  injects charge to  $C$ , the total charge at node  $X$  cannot change!

- Let us process total **charge** at node  $X$  instead of looking at voltage across  $C$ .

$$Q_X = -CV_{in} + \Delta Q_2/2$$

- The charge can be processed in two ways
  - Open-loop
  - Closed-loop

# Open-Loop Charge Processing



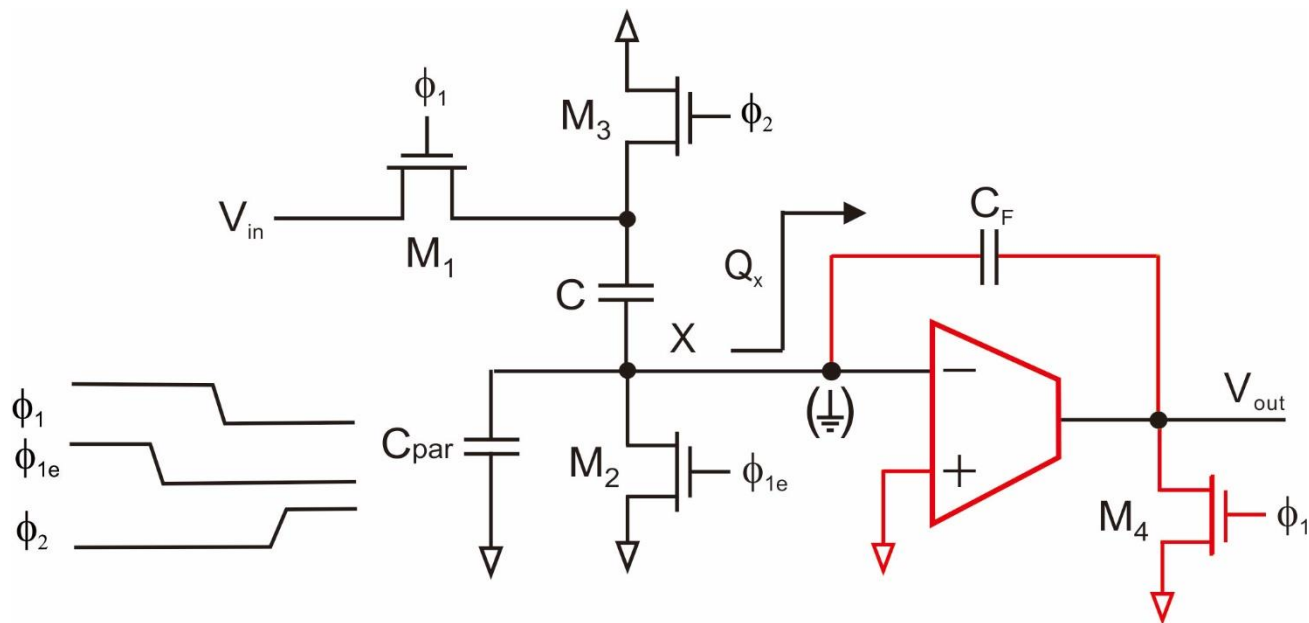
- During  $\phi_1$ ,  $V_{in}$  is sampled at X.  $Q_x$  is linearly proportional to  $V_{in}$ .
- Due to charge conservation,  $Q_x$  does not change when  $M_3$  is on  $\rightarrow$

$$Q_x = -CV_{in} + \Delta Q_2/2 = (C + C_{par})V_x$$

$$V_{out} = V_x = -\frac{C}{C + C_{par}}V_{in} + \frac{\Delta Q_2/2}{C + C_{par}}$$

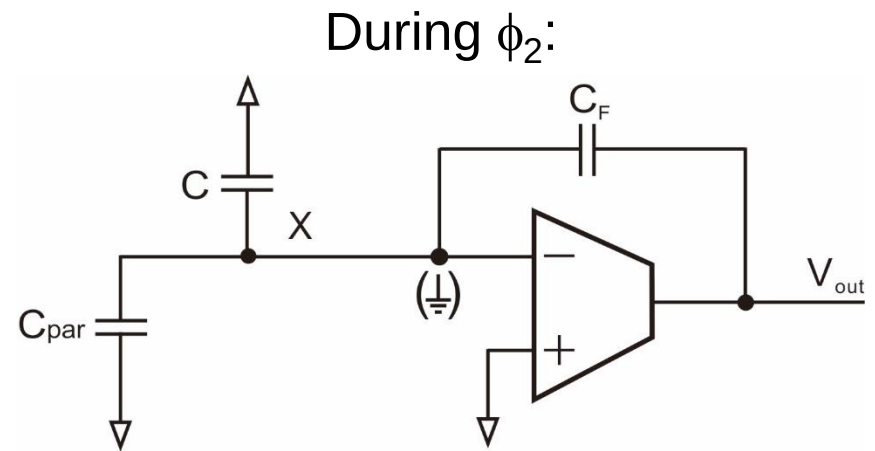
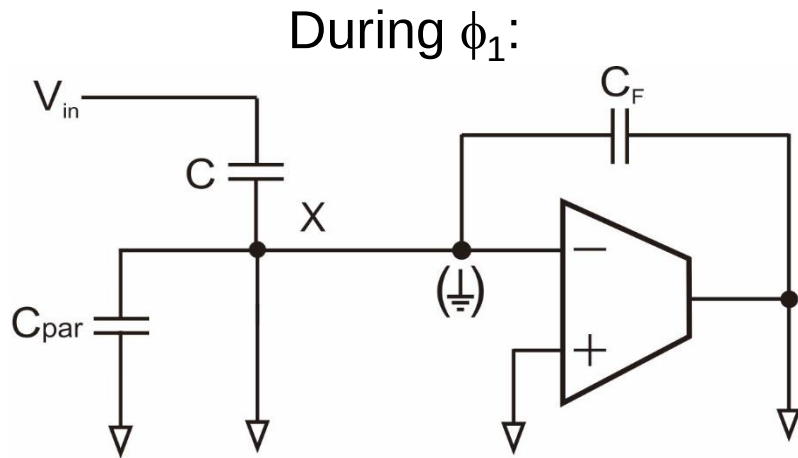
- Signal flow:  $V_{in} \rightarrow Q_x \rightarrow V_x$
- Remaining drawback
  - $C_{par}$  includes transistor parasitic cap. It is nonlinear and will introduce distortion.

# Closed-Loop Charge Processing



- The opamp forces voltage at node  $X$  to “zero”
  - Means that charge at node  $X$  must redistribute onto feedback capacitor  $C_F$

# Charge Conservation Analysis



Charge at node X during when  $M_2$  is off:

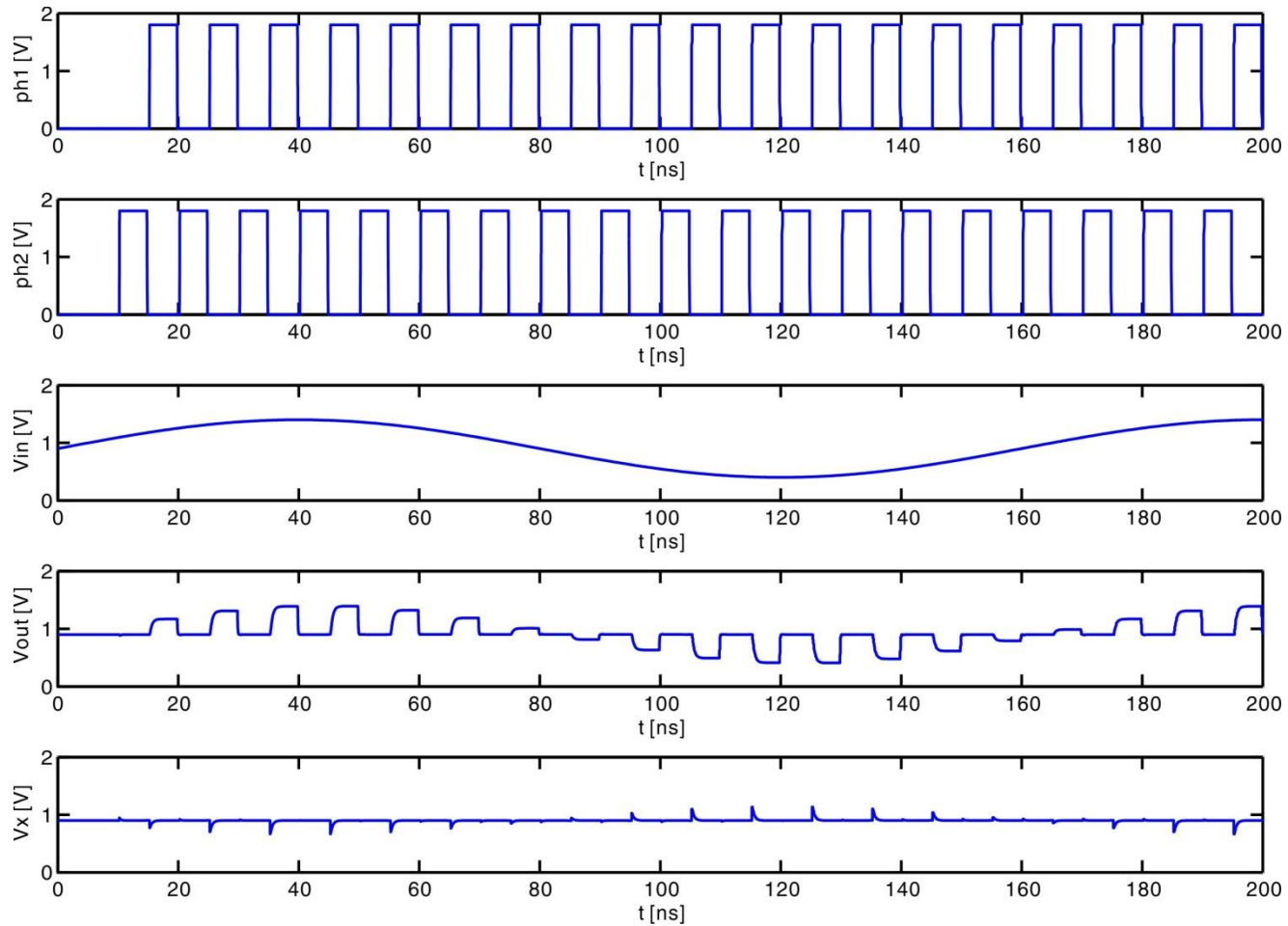
$$Q_{x1} = -CV_{in} + \Delta Q_2/2$$

Charge at node X during  $\phi_2$ :

$$Q_{x2} = -C_F V_{out}$$

$$Q_{x1} = Q_{x2} \quad \Rightarrow \quad V_{out} = \frac{C}{C_F} V_{in} + \frac{-\Delta Q_2}{2C_F}$$

# Behavioral Simulation

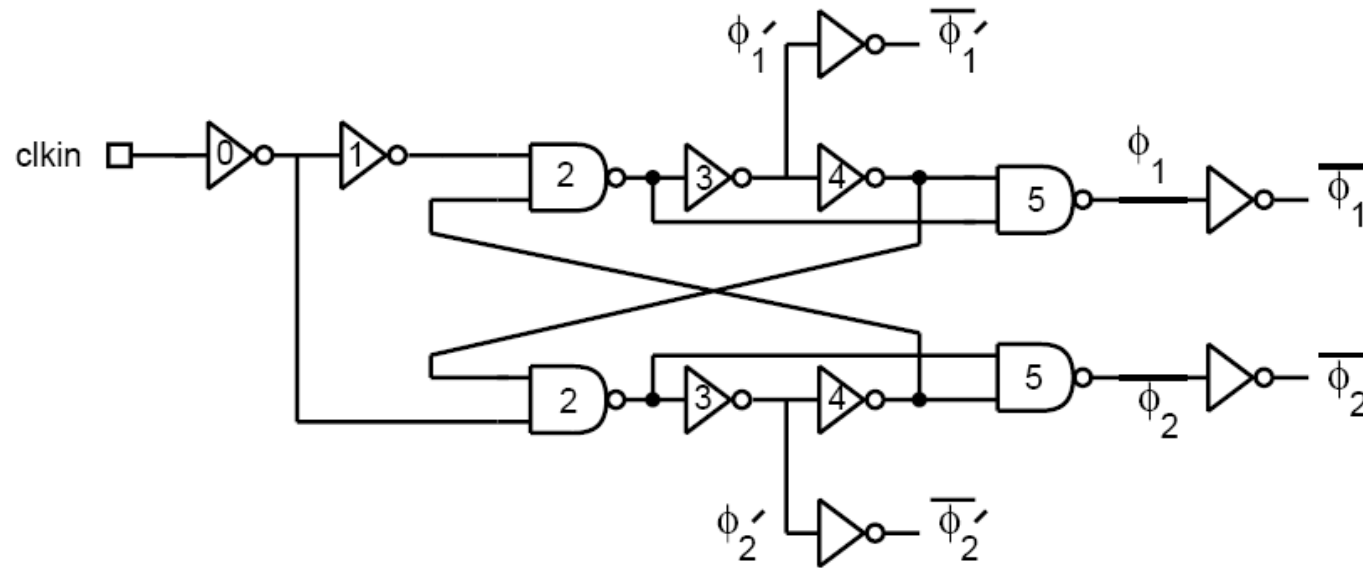


# Comments

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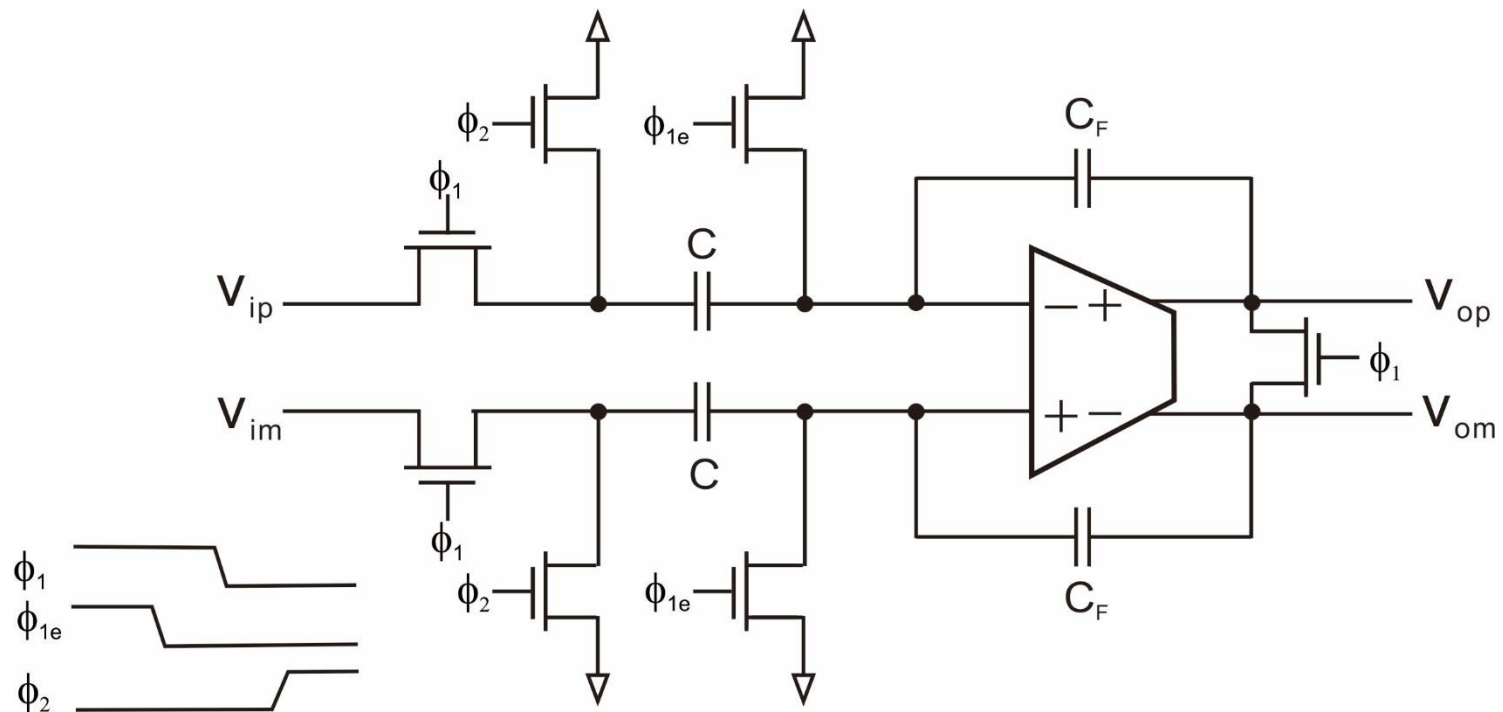
- $C_F$  is highly linear (more later)
- Gain can be provided.
- Offset term can be easily removed by going differential.
- Linearity can be limited by opamp (offset, insufficient gain, nonlinear gain)

# Clock Generation



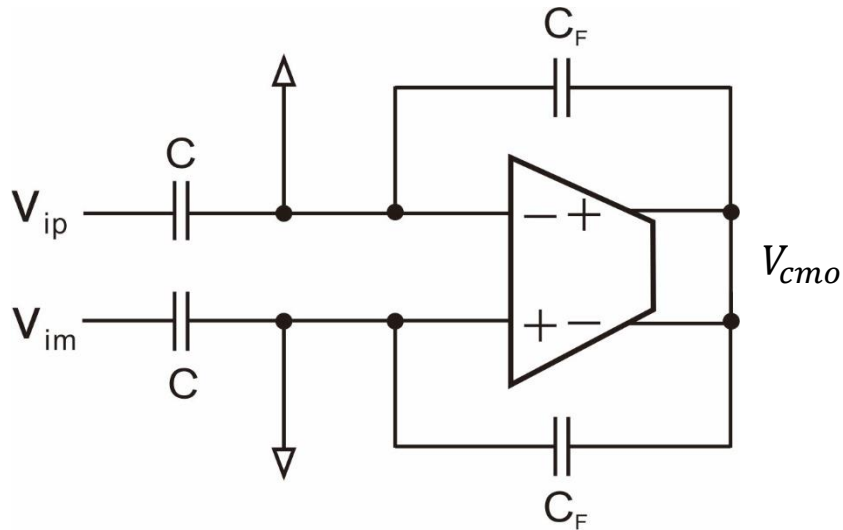
[A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits," PhD Thesis, UC Berkeley, 1999]

# Fully Differential Circuit



# Analysis (1)

During  $\phi_1$

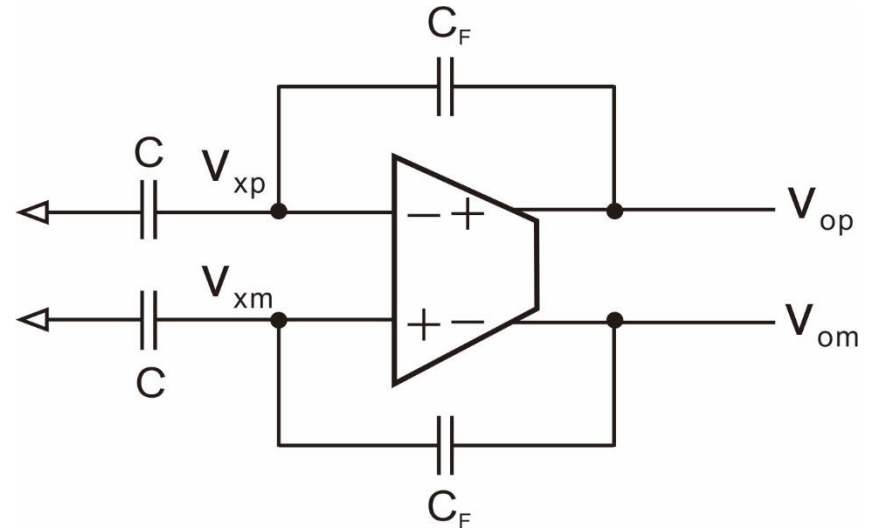


$$Q_{1m} = -CV_{ip} - C_F V_{cmo} + \Delta Q$$

$$Q_{1p} = -CV_{im} - C_F V_{cmo} + \Delta Q$$

$\longleftrightarrow$

During  $\phi_2$



$$Q_{2m} = CV_{xm} + C_F(V_{xm} - V_{om}) = Q_{1m} \quad (1)$$

$$Q_{2p} = CV_{xp} + C_F(V_{xp} - V_{op}) = Q_{1p} \quad (2)$$

$$V_{xm} = V_{xp}$$

# Analysis (2)

---

- Subtracting 1) and 2), we obtain:

$$V_{out} = V_{op} - V_{om} = \frac{C}{C_F} (V_{ip} - V_{im}) = \frac{C}{C_F} V_{in}$$

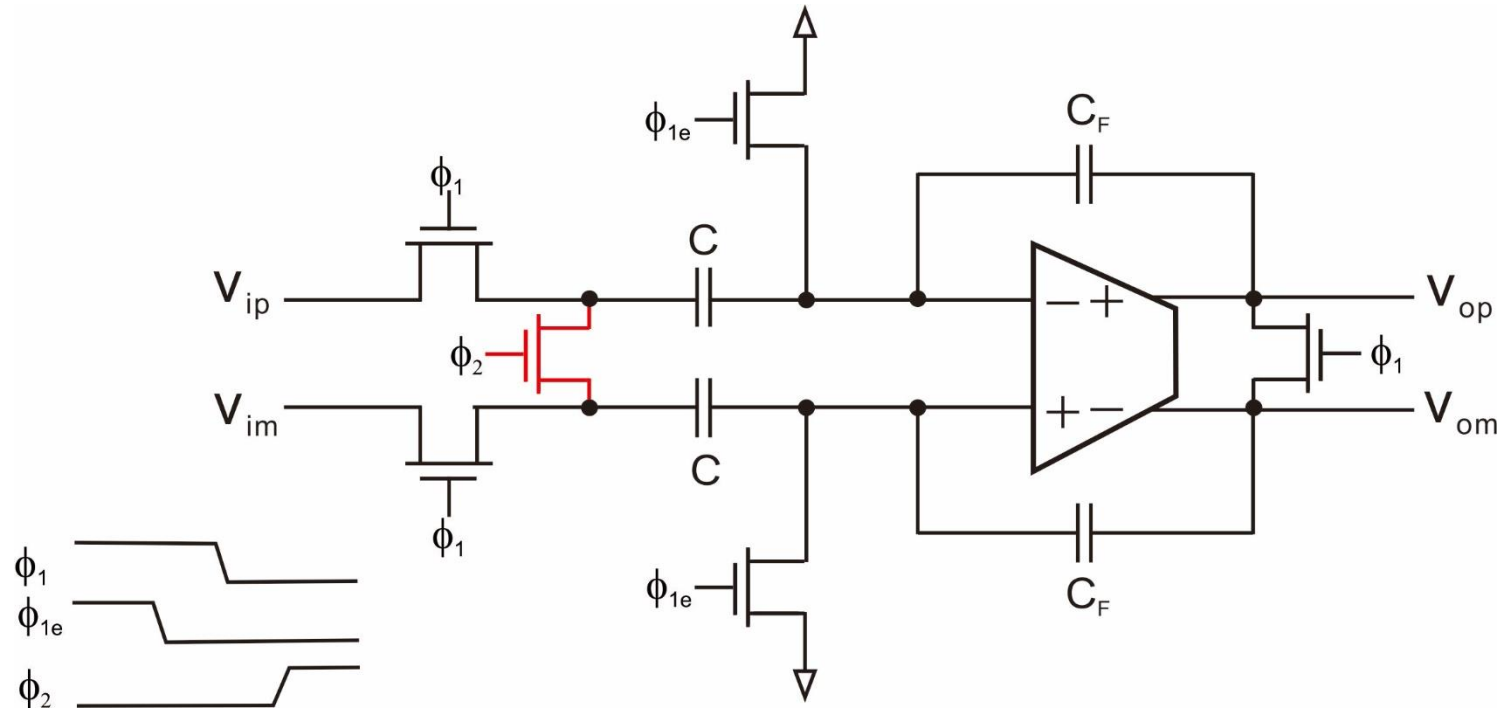
- Adding 1) and 2), we obtain:

$$\begin{aligned} -C(V_{ip} + V_{im}) - 2C_F V_{cmo} + 2\Delta Q &= (C + C_F)(V_{xp} + V_{xm}) - C_F(V_{op} + V_{om}) \\ -C V_{cmi} + \Delta Q &= (C + C_F) V_{cmx} \end{aligned}$$

$$\Rightarrow V_{cmx} = -\frac{C}{C + C_F} V_{cmi} + \frac{\Delta Q}{C + C_F}$$

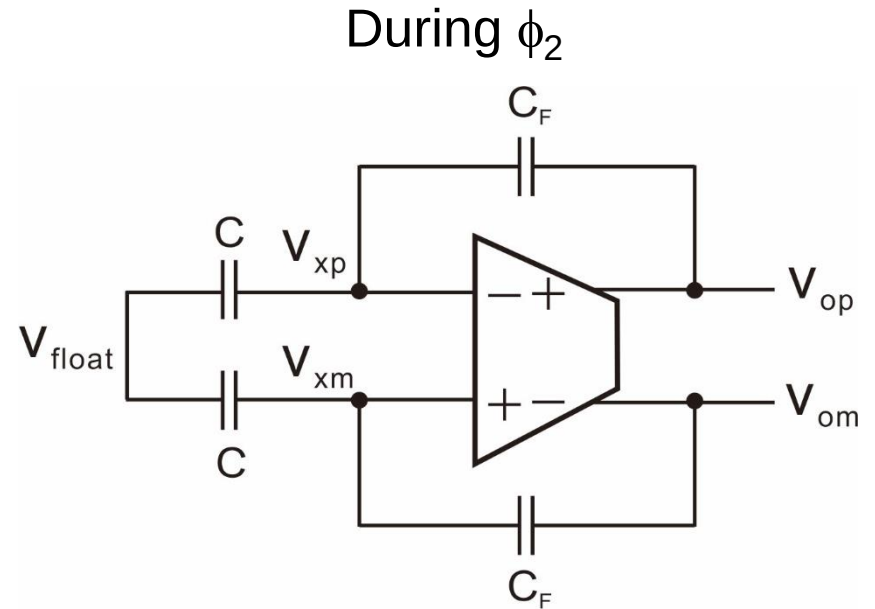
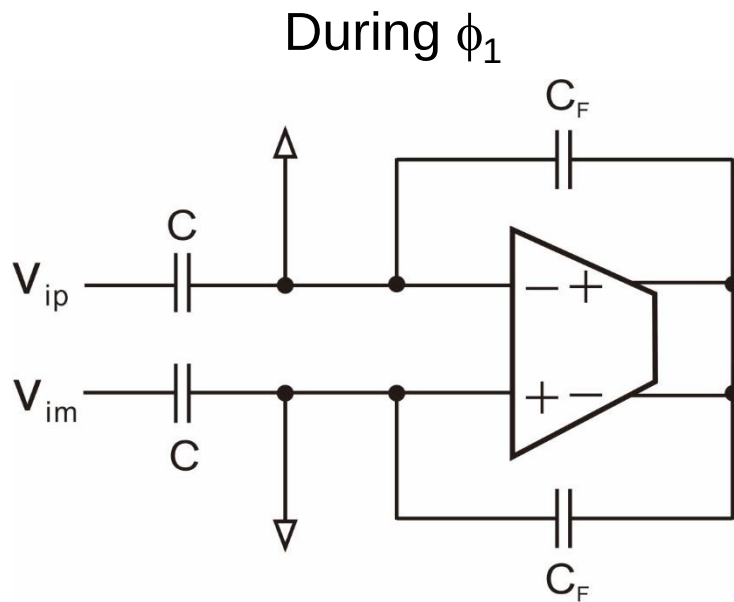
- $V_{cmi}$  (input common mode) shows up at the opamp input
  - Opamp must have wide input common mode range
  - Opamp must have good CMRR

# T/H with Common Mode Cancellation



- Shorting switch allows to re-distribute only differential charge
- Common mode at opamp input becomes independent of  $V_{cmi}$

# Analysis (1)



- Charge conservation at the left plate of C:

$$(V_{ip} + V_{im})C = (V_{float} - V_{xp})C + (V_{float} - V_{xm})C$$

$$V_{cmi} = V_{float} - V_{cmx}$$

# Analysis (2)

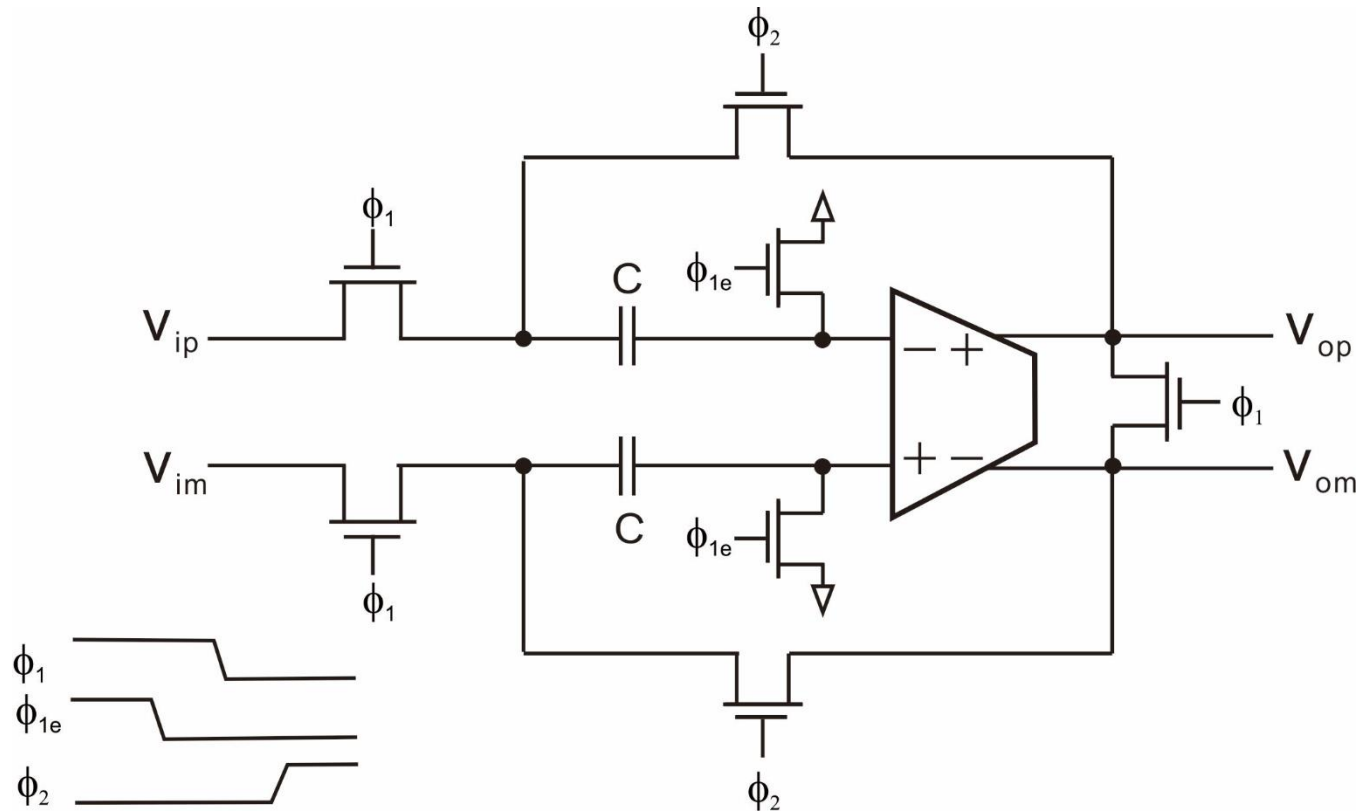
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- Common mode charge conservation at the opamp input node:

$$\begin{aligned} -V_{cmi}C - V_{cmo}C_F &= -(V_{float} - V_{cmx})C - (V_{cmo} - V_{cmx})C_F \\ -V_{cmi}C &= -[(V_{cmi} + V_{cmx}) - V_{cmx}]C + V_{cmx}C_F \\ 0 &= V_{cmx} \end{aligned}$$

- Thus, opamp input common mode  $V_{cmx}$  is independent of
  - Input common mode  $V_{cmi}$
  - Output common mode  $V_{cmo}$ 
    - Usually set by common mode feedback circuit

# Flip-Around T/H

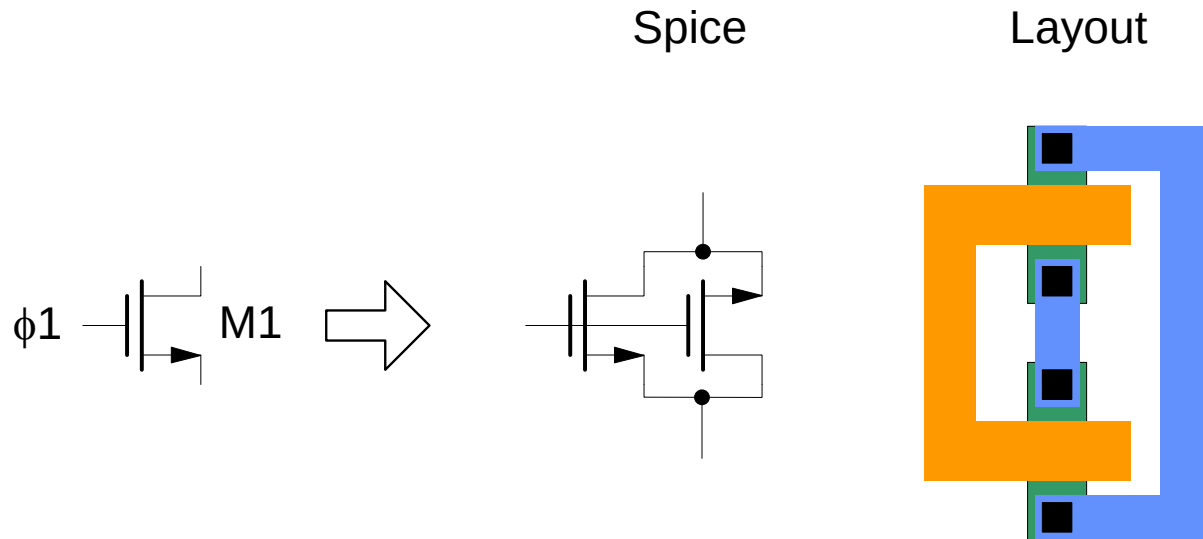


- The same cap used for both sampling and opamp feedback
  - Larger feedback factor (lower noise, higher speed) 😊
  - OTA is subjected to input common mode variations ☹️

The diagram shows a fully differential folded-cascode operational amplifier. It features a differential pair of NMOS transistors (M<sub>1</sub>, M<sub>2</sub>) with their sources connected to a common source node. The gates of M<sub>1</sub> and M<sub>2</sub> are driven by a differential-mode input signal  $V_{ip}$  and a common-mode input signal  $V_{im}$ . The gates of M<sub>1</sub> and M<sub>2</sub> are also biased by a common-mode gate voltage  $\phi_1$ . The drains of M<sub>1</sub> and M<sub>2</sub> are connected to a differential pair of PMOS transistors (M<sub>3</sub>, M<sub>4</sub>) through a load capacitor  $C_F$ . The gates of M<sub>3</sub> and M<sub>4</sub> are biased by a common-mode gate voltage  $\phi_1$ . The drains of M<sub>3</sub> and M<sub>4</sub> are connected to a differential-mode output signal  $V_{op}$  and a common-mode output signal  $V_{om}$ . The gates of M<sub>3</sub> and M<sub>4</sub> are also biased by a common-mode gate voltage  $\phi_1$ . The gates of M<sub>1</sub> and M<sub>2</sub> are biased by a common-mode gate voltage  $\phi_1$ . The gates of M<sub>3</sub> and M<sub>4</sub> are biased by a common-mode gate voltage  $\phi_1$ . The gates of M<sub>1</sub> and M<sub>2</sub> are biased by a common-mode gate voltage  $\phi_1$ . The gates of M<sub>3</sub> and M<sub>4</sub> are biased by a common-mode gate voltage  $\phi_1$ .

- 64

# Schematic and Layout of $M_1$



From B. Murmann, EE 315, Ch. 5

- $M_1$  is usually implemented using anti-parallel devices to ensure symmetry
  - Drain and source are not necessarily symmetric in circuit models
  - Drain and source are not symmetric due to process artifacts

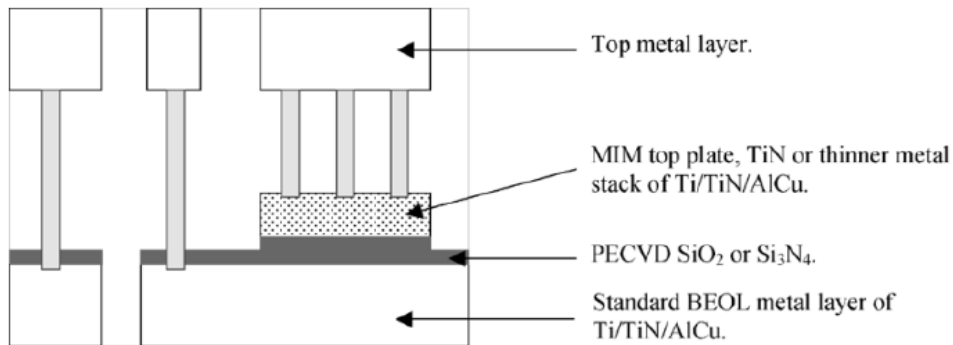
# What Ultimately Limits Linearity?

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- Bottom-plate sampling plus clock bootstrapping solve majority nonlinearity problems. We can achieve 80~100dB linearity up to a few hundreds of MHz.
- Limitation of clock bootstrapping:  $R_{\text{top,SW}}$  varies with  $V_{\text{in}}$  due to **back-gate effects** and **parasitic capacitances** (especially at high frequency)
  - In the track phase, it causes **nonlinear filter response**
    - Bottom-plate sampling does not help.
  - In the hold phase, it causes **input dependent charge injection** because the charge splitting depends on the impedance
  - We can mitigate this issue by reducing  $R_{\text{top,SW}}$
- Limitation of bottom-plate sampling
  - Open-loop charge transfer: nonlinear parasitic cap
  - Close-loop charge transfer: opamp nonlinearity

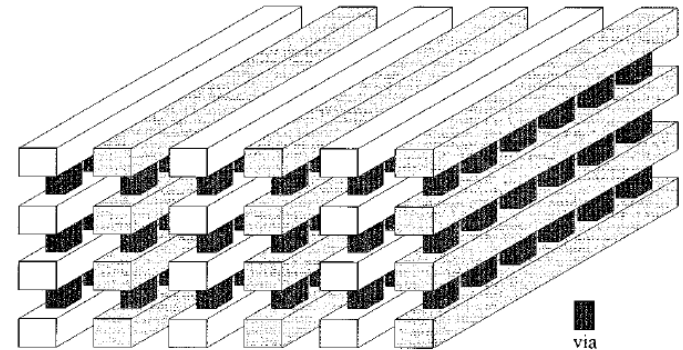
# Capacitors

## Metal-Insulator-Metal (MIM)



[Ng, Trans. Electron Dev., 7/2005]

## Metal-On-Metal (MOM)

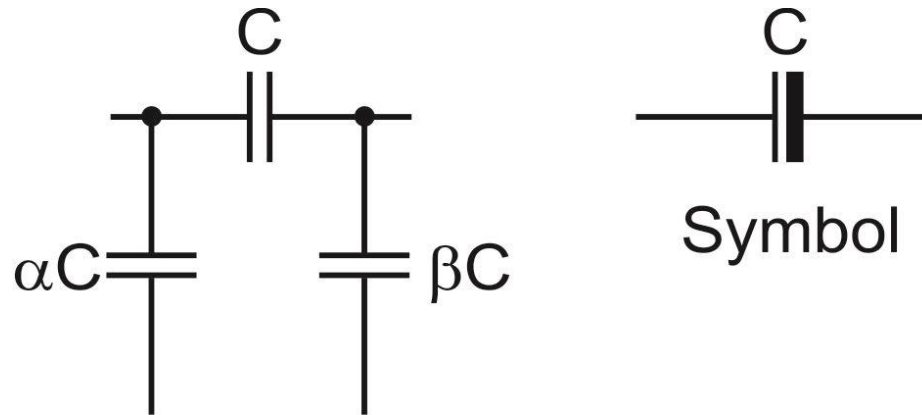


[Aparicio, JSSC 3/2002]

- Both MIM and MOM capacitors have good electrical properties
  - Series and parallel resistances are usually not a concern
  - Mostly worry about parasitic caps
- MOM caps are preferred in advanced processes due to higher density.
  - The density increases as process scales.
- MIM cap density is usually 1-2 fF/ $\mu\text{m}^2$ 
  - For 1 fF/ $\mu\text{m}^2$ , a 10 pF capacitor occupies 100 $\mu\text{m}$  x 100 $\mu\text{m}$

# Plate Parasitics

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- The two plates of a real capacitor are usually unsymmetrical
  - We usually use the fat plate to indicate the plate that has larger parasitic capacitance.
- For a MIM capacitor, the plate that is away from the substrate has smaller parasitic capacitance (typical value:  $\alpha=1\%$ ,  $\beta=10\%$ ).
- For a MOM capacitor,  $\alpha$  and  $\beta$  can vary.
- We can change the effective  $\alpha$  and  $\beta$  through layout design.

# Proper Connection of Capacitors

- The more ideal plate should face the virtual ground nodes
  - Avoid reduction of feedback factor
  - Reduce potential noise coupling

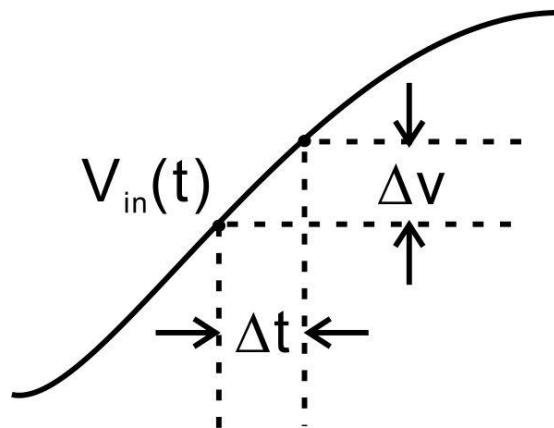
# Nonidealities

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- Switch resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities

# Sampling Clock Jitter

- Clock jitter introduces sampling errors.
  - Its magnitude is proportional to the variation of  $V_{in}$ .



$$\Delta V_{in} \approx \frac{dV_{in}}{dt} \cdot \Delta t$$

- Let us consider a sine wave input signal and assume  $\Delta t$  follows normal distribution with zero mean and standard deviation  $\sigma_t$

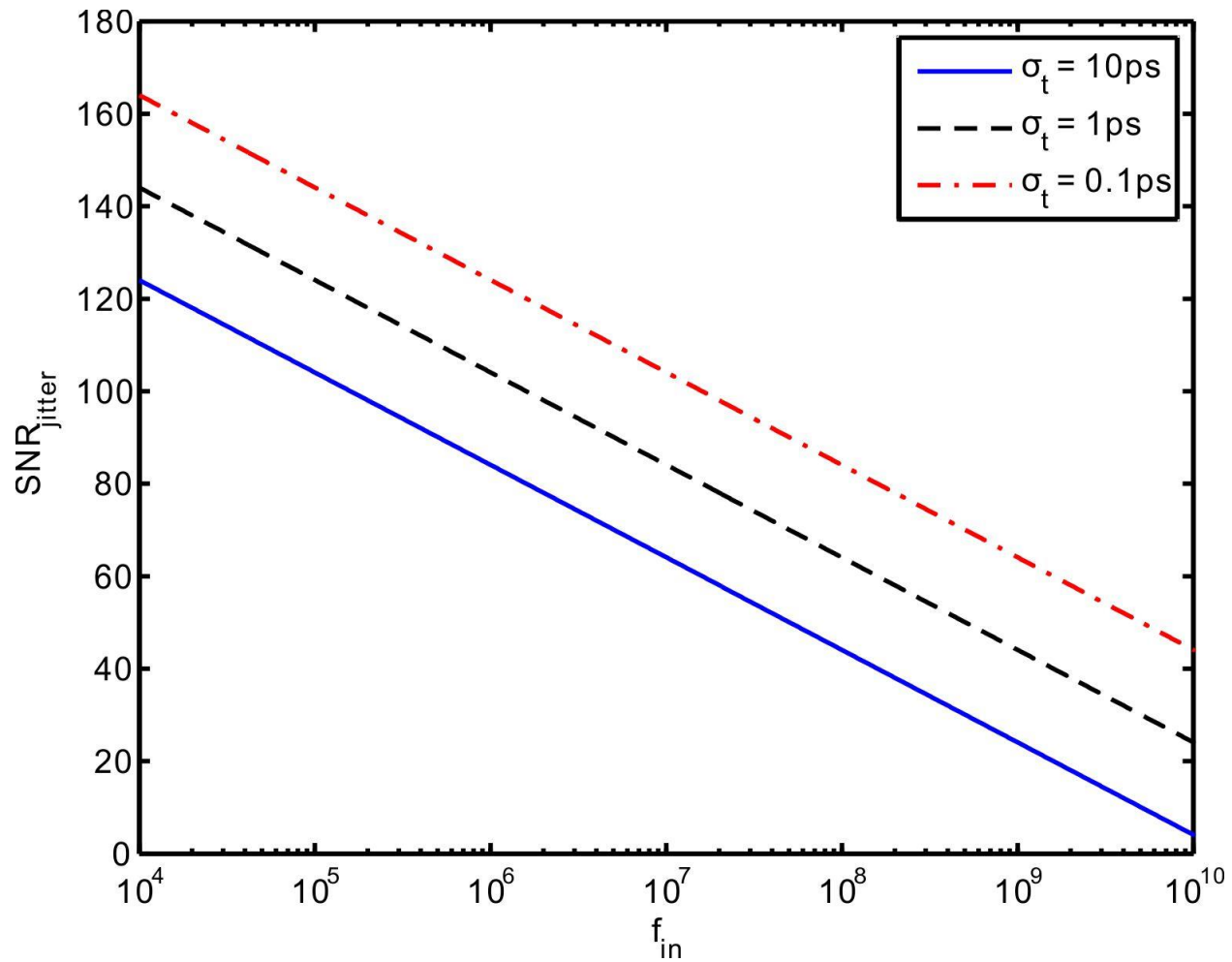
# Analysis

$$\begin{aligned} E\{\Delta V_{in}^2\} &\cong E\left\{\left(\frac{dV_{in}}{dt}\right)^2 \cdot \Delta t^2\right\} = E\left\{\left(\frac{dV_{in}}{dt}\right)^2\right\} \cdot E\{\Delta t^2\} \\ &\cong E\left\{\left(\frac{d}{dt} A \cos[2\pi \cdot f_{in} \cdot t]\right)^2\right\} \cdot \sigma_t^2 \cong \frac{1}{2} (2\pi \cdot A \cdot f_{in})^2 \cdot \sigma_t^2 \end{aligned}$$

$$\boxed{\text{SNR}_{\text{jitter}} [\text{dB}] \cong 10 \cdot \log \left[ \frac{\frac{1}{2} A^2}{\frac{1}{2} (2\pi \cdot A \cdot f_{in} \cdot \sigma_t)^2} \right] = 20 \cdot \log \left[ \frac{1}{2\pi \cdot f_{in} \cdot \sigma_t} \right]}$$

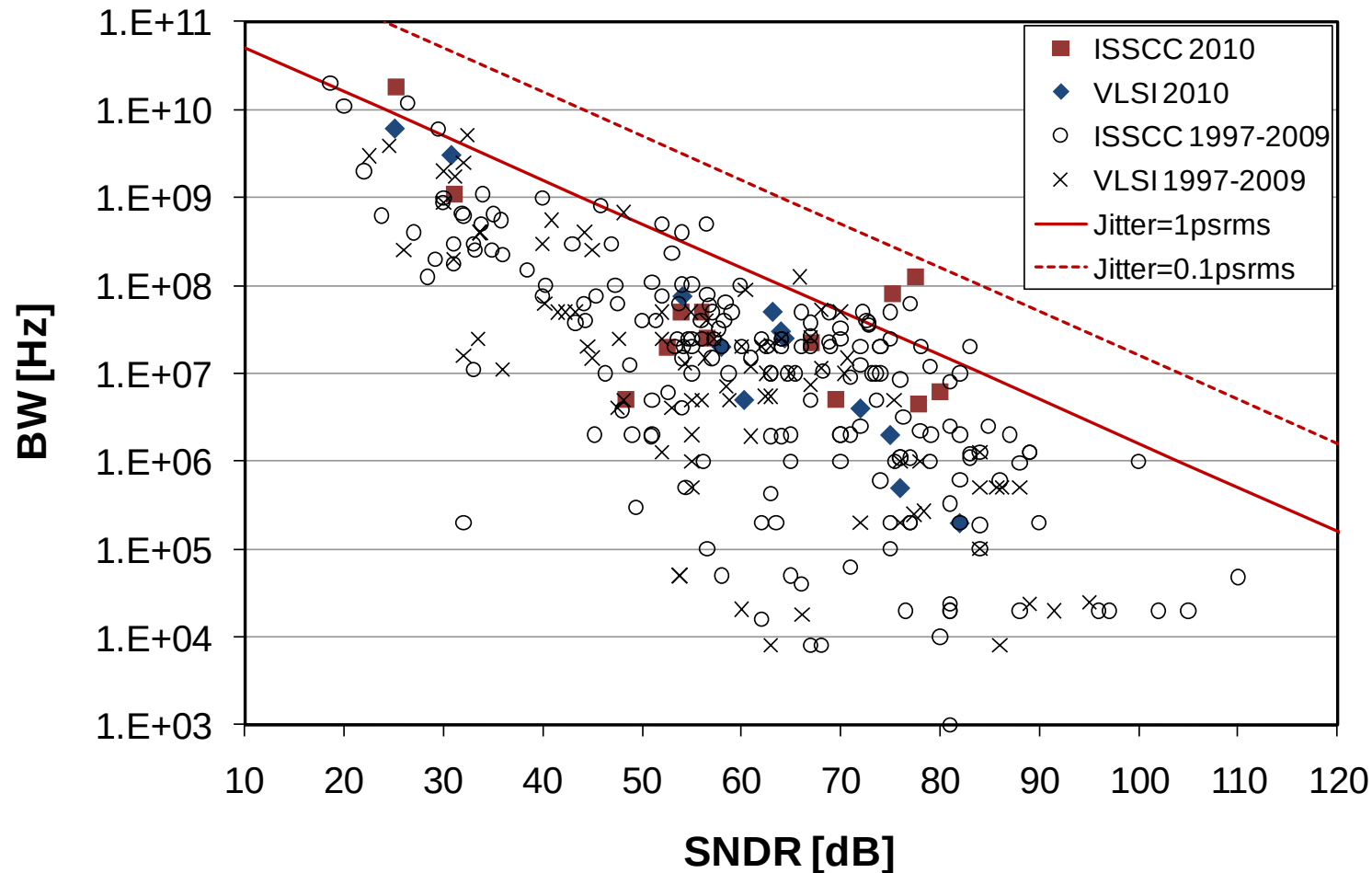
- Note that this jitter limited SNR does not depend on signal amplitude!
- The jitter induced error increases as signal amplitude increases.
  - In testing, if SNR remains constant for various signal amplitudes, it is highly possible that the SNR is limited by clock jitter.

# Result



# ADC Performance Survey

From B. Murmann, EE 315, Ch. 5

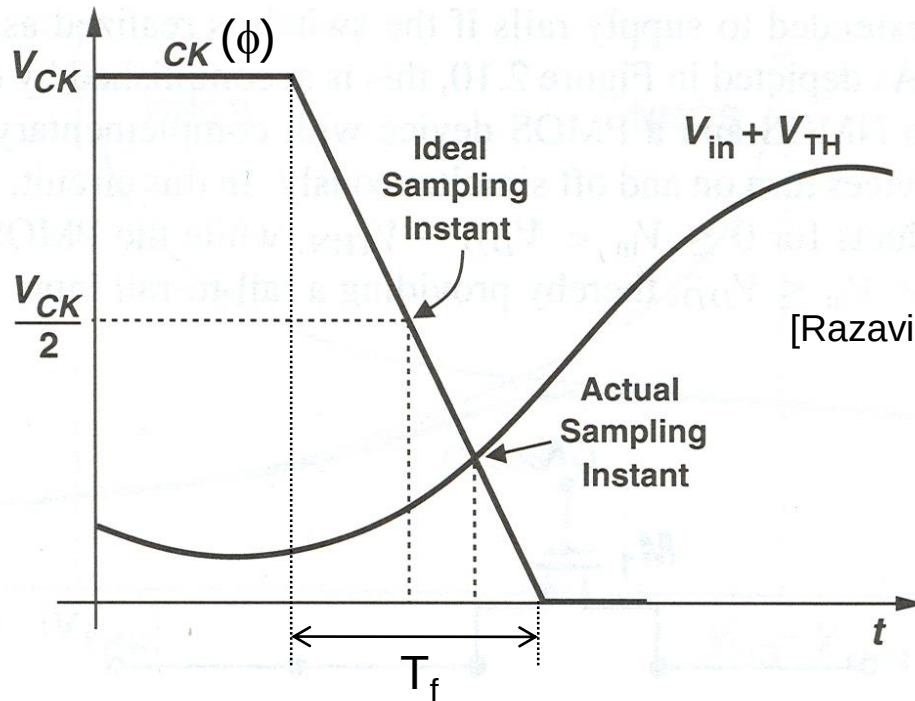


# Nonidealities

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- Switch resistance
- Switch capacitance
- Sampling clock jitter
- Other nonidealities
  - Signal dependent sampling instants
  - Hold mode feed-through
  - Hold mode leakage

# Signal Dependent Sampling Instant (1)



[Razavi, Data Conversion System Design, p.17]

- Must make  $T_f$  much smaller than maximum  $dV_{in}/dt$ 
  - This means that we are in the fast-gating scenario.

# Signal Dependent Sampling Instant (2)

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- Distortion analysis result (see Yu, TCAS II, 2/1999]

$$HD_3 = \frac{3}{8} \left( \frac{A}{V_{CLK}} \omega T_F \right)^2$$

- Example #1:  $V_{CLK} = 1.8V$ ,  $A = 0.5V$ ,  $T_f = 100ps$ ,  $\omega = 2\pi \times 100MHz$

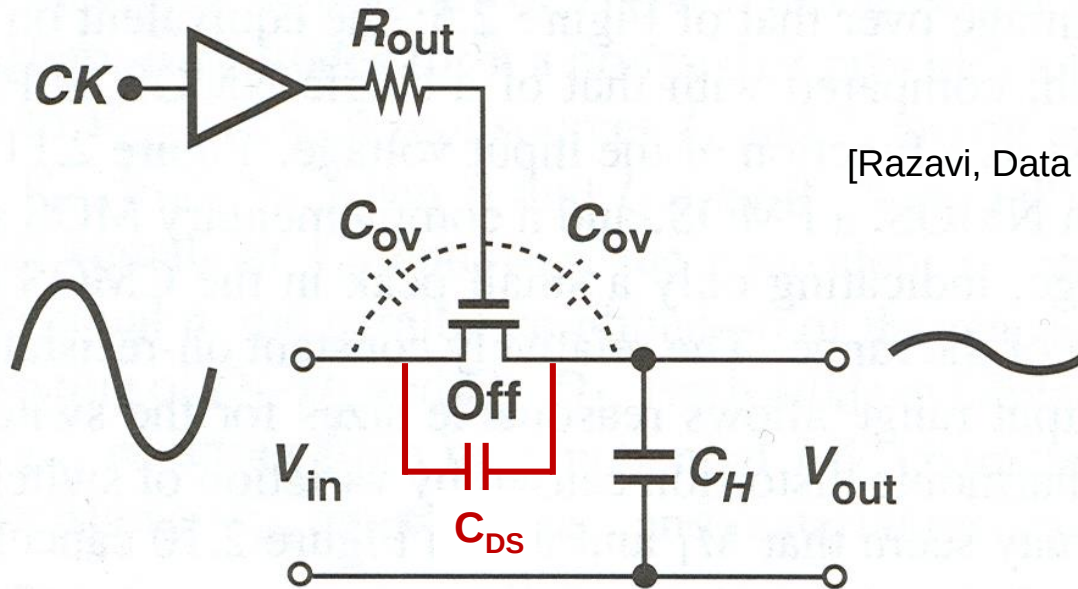
$$HD_3 = \frac{3}{8} \left( \frac{0.5}{1.8} 2\pi \cdot 100e6 \cdot 100e-12 \right)^2 = -79dB$$

- Example #2:  $V_{CLK} = 1.8V$ ,  $A = 0.5V$ ,  $T_f = 100ps$ ,  $\omega = 2\pi \times 1GHz$

$$HD_3 = \frac{3}{8} \left( \frac{0.5}{1.8} 2\pi \cdot 1e9 \cdot 100e-12 \right)^2 = -39dB$$

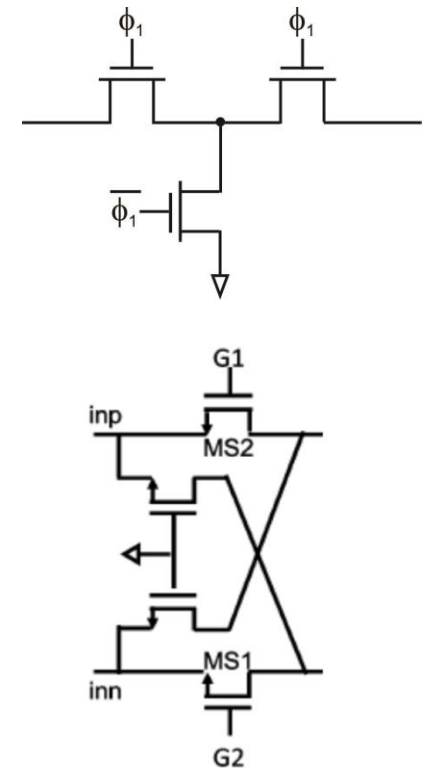
- Short  $T_f$  is a must for sampling high frequency signal
  - Note that a realistic  $T_f$  should be set in simulation

# Hold Mode Feedthrough



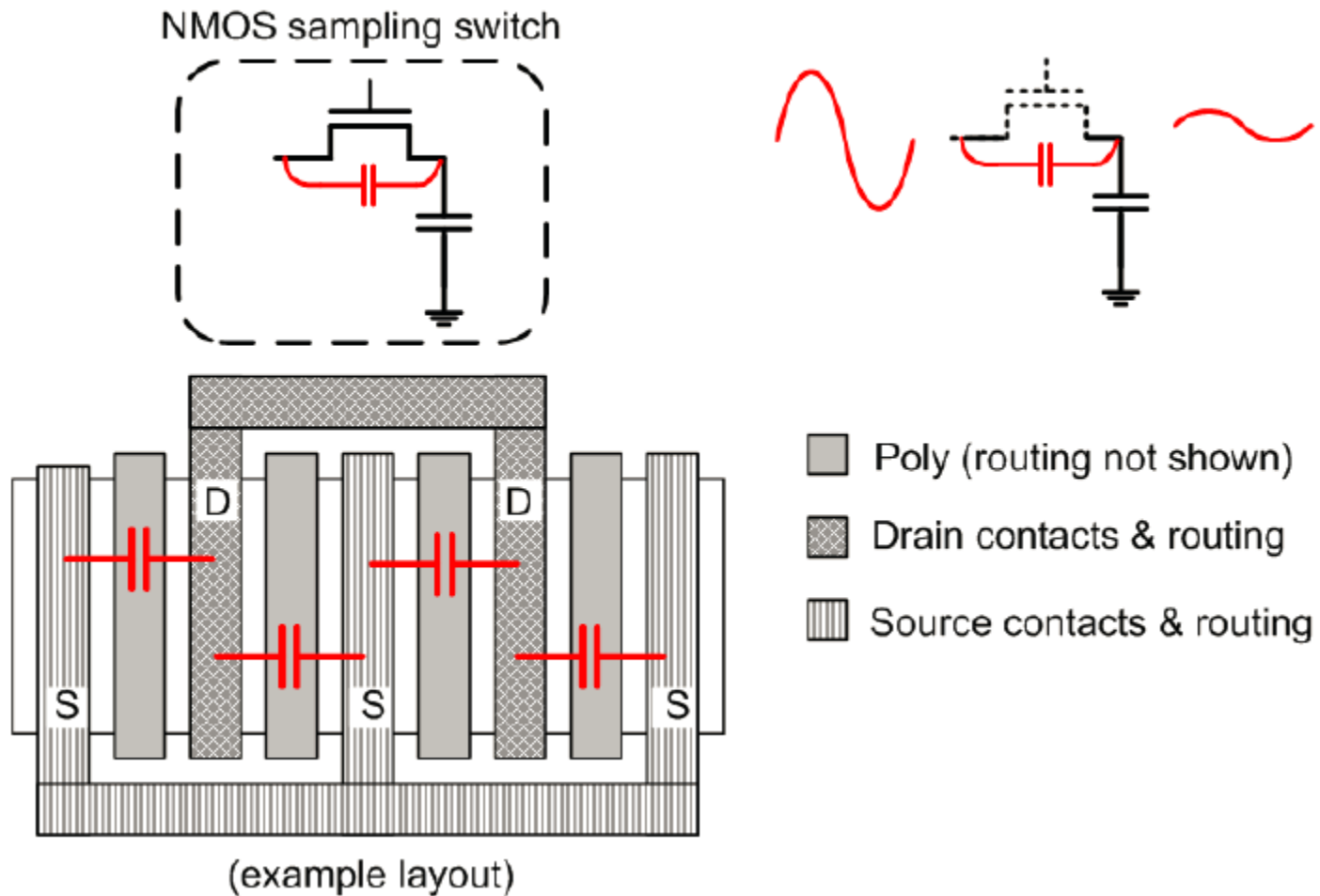
[Razavi, Data Conversion System Design, p.17]

- Make  $R_{out}$  as small as possible
- Use T-switch when hold-mode feedthrough is a problem (for low-speed)
- Use cross coupling for cancellation (for high-speed)
- Minimize  $C_{DS}$  in layout



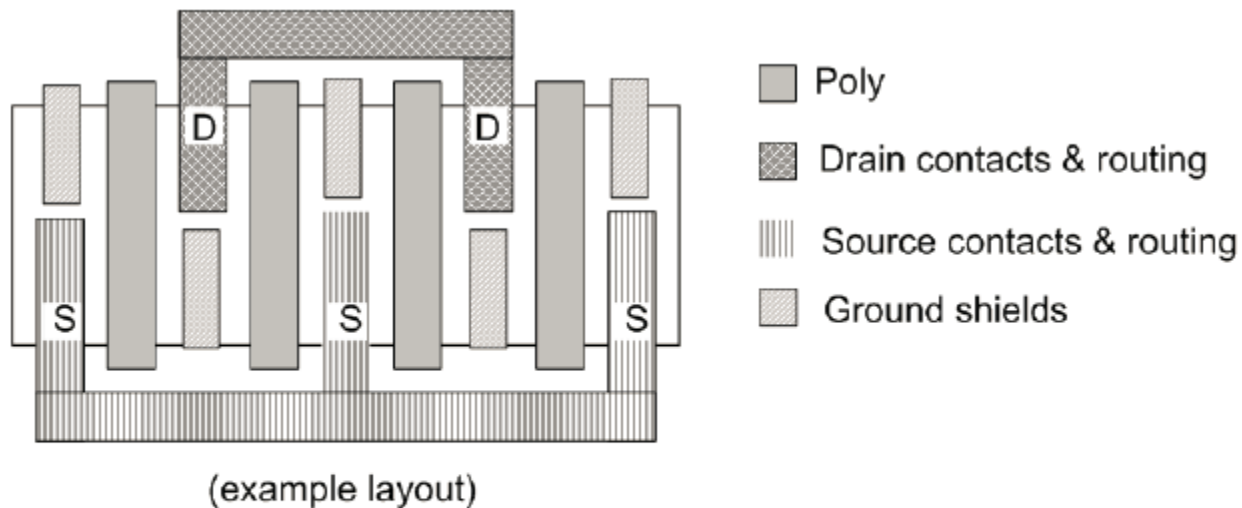
# Minimize $C_{DS}$ (1)

[Manar El-Chammas, ISSCC 2015, 15.8 slide]



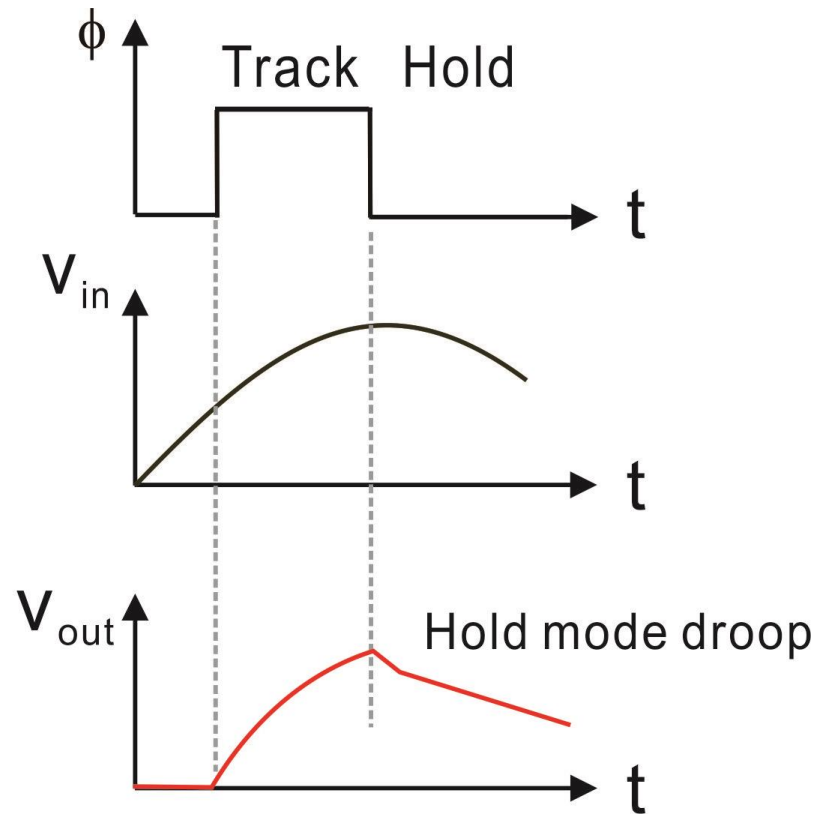
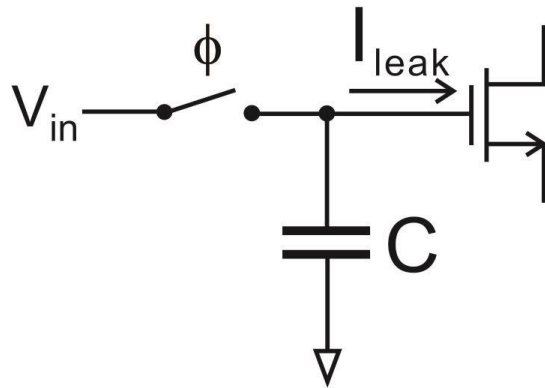
# Minimize $C_{DS}$ (2)

[Manar El-Chammas, ISSCC 2015, 15.8 slide]

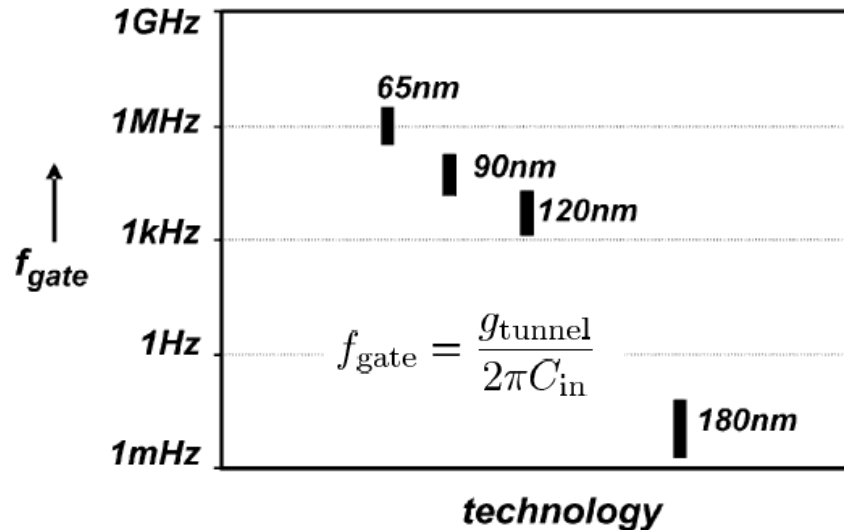


- Reduce contacts  $\rightarrow$  capacitance cut  $\sim 15X$
- Add ground shields  $\rightarrow > 30X$  reduction

# Hold Mode Leakage



# Gate Leakage Data



A. Annema, et al., "Analog circuits in ultra-deep-submicron CMOS," *IEEE J. Solid-State Circuits*, pp. 132-143, Jan. 2005.

- $f_{gate}$  indicates how fast charge leaks
- Gate leakage depends on dielectric thickness and gate voltage
- The use of high-K dielectric ( $H_f$ ) reduces gate leakage
- Gate leakage is more troublesome for low speed operation!
  - Avoid holding charges for a long time